Foreign discrimination, Business Lobbying, and the Unraveling of the U.S.-China Free Trade Consensus

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Last revised: October 14, 2025

Abstract

How does corporate lobbying contribute to the unraveling of U.S. elite support for free trade with China? Existing models of legislators' trade positions largely attribute firm influence to their material production profiles. I argue that foreign discrimination—host-state policies and institutional environments that systematically disadvantage foreign firms—plays a central role in shaping firm interests and lobbying strategies. I evaluate this argument in the context of U.S.-China trade relations between 2001 and 2022. Based on financial filings from all publicly traded U.S. firms, I leverage large language models to classify firm-level interests in global value chain (GVC) engagement and experiences of discriminatory policies related to China. I construct novel measures of firm's lobby-transmitted interests based on lobby network data and estimate their effects on legislative rhetoric. I present three main findings. First, firms that cite discriminatory policies in China increase trade-critical rhetoric among the legislators they lobby. Second, firms that benefit from GVC engagement with China, particular through import sourcing, are able to mitigate some trade criticism. Finally, concerns over intellectual property (IP) have emerged as a new driver of trade criticism during the Trump and Biden administrations, reflecting the intensifying conflict over high-tech sectors between the two nations. These findings underscore the need to treat firm interests as endogenous to host-state policies and institutional contexts, and reveal how GVC-engaged firms can be fragmented and mobilize new anti-trade coalitions.

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1 Introduction

The rise of global value chains (GVCs) has blurred the boundaries of nations in trade, as countries and firms now participate in complex networks of intermediate trade and foreign investment beyond simple exports and imports. The boundaries of trade politics are similarly blurred: host-state policies, even when designed to address domestic concerns, can reverberate through these value chains, affecting foreign firms embedded within them and, ultimately, shaping home-state policy responses. The trade politics literature, however, has predominantly focused on how domestic cleavages shape policymakers' preferences on free trade, whether along factors of production (labor versus capital; Rogowski 1989), across industries (import-competing versus export-oriented; Hiscox 2002), or within industries (between large, productive and unproductive firms; Kim and Osgood 2019). A common assumption is that industries' or firms' interests are determined by their material "production profiles" (Gourevitch 1986), with host-state policies or institutional environments treated as exogenous or background conditions.

Assuming away host-state politics makes it difficult to account for the temporal shifts in firms' attitudes toward the trade regime. Consider American technology giants' attempts to access the Chinese market. In 2006, Google officially launched their search service in China while attempting to comply with stringent government censorship requirements. Four years later, however, it announced a "new approach to China" and cited sophisticated cyberattacks on its infrastructure as the reason for discontinuing its search operations.[1] Similarly, in 2012 Facebook stated that it would "continue to evaluate entering China" as a priority, and CEO Mark Zuckerberg sought to deepen personal ties with top Chinese leaders. Yet as the prospect of market access dwindled, Zuckerberg later adopted a more confrontational stance, testifying before Congress in 2020 that China's internet system is antithetical to American values.[2] While traditional political-economy models could correctly predict the initial enthusiasm among U.S. high-tech service firms toward the Chinese market, they are relatively muted about the subsequent shift in their attitudes.

1. https://googleblog.blogspot.com/2010/01/new-approach-to-china.html
2. https://www.sec.gov/Archives/edgar/data/1326801/000132680113000003/fb-12312012x10k.htm; https://docs.house.gov/meetings/ju/ju05/20200729/110883/hhrg-116-ju05-wstate-zuckerbergm-20200729.pdf

This paper treats firm interests as endogenous to host-state policy and institutions, showing how foreign discrimination shapes lobbying patterns and contributes to the rise of new anti-trade coalitions. It is built on the premise that market access and production efficiency—two principal reasons for firms to engage in international trade or investment—are parameters over which the host state has substantive control. It does so through not just tariffs but also a host of behind-the-border policies, a central focus of the literature on "deep" trade agreements (Büthe and Milner 2014; Dür, Baccini, and Elsig 2014; Baccini, Pinto, and Weymouth 2017). The discriminatory impact of host-state policies on foreign firms is unevenly distributed across and within industries and over time.

Firms lobby politicians to provide private information about instances of foreign discrimination, subsidizing the latter's legislative efforts on the one hand (Hall and Deardorff 2006), and avoiding direct public criticism that could invite political retaliation from the host state on the other (Cha 2023; Lim and Ferguson 2022). Legislators have stronger incentives to embrace trade criticism that highlights foreign discrimination because it activates sociotropic concerns among voters, such as reciprocity, fairness, and in-group favoritism (Mutz and Kim 2017; Chilton, Milner, and Tingley 2020), thereby reducing the likelihood of electoral backlash. Trade criticism from Congress can exert direct pressure on the host government or indirectly by shaping the executive branch's agenda in bilateral negotiations.

Empirically, I focus on the U.S.–China trade relations between 2001 and 2022. I introduce a novel text-as-data approach to measure firm-level interests in China and estimate firms' lobbying influence. I compile 58,960 financial filings submitted to the U.S. Securities and Exchange Commission (SEC) by 11,300 unique publicly traded companies that mention China. Leveraging open-source large language models (LLMs), I annotate every firm's discussion of China with key categories of firm interests—export, import, foreign direct investment (FDI), complaints about discriminatory policies, and about intellectual property (IP) infringement—in a transparent, reproducible, and efficient manner. I then construct original measures of firms' *lobby-transmitted interests* to legislators based on lobbying network data (I. S. Kim 2018). Finally, I estimate how these firms' lobby-transmitted interests shape legislators' rhetoric on trade with China using the Congressional Record covering the 107th–117th Congresses.

I present three main findings. First, firms that cite discriminatory policies in China promote trade-critical rhetoric among the legislators they lobby, particularly on China's "unfair" trade practices. The relationship is robust to numerous alternative explanations, including district-level economic characteristics, China's import shocks, politicians' military hawkishness and potential contemporaneous bias. Second, firms that benefit from GVC engagement with China, particularly through import sourcing, are able to mitigate some trade criticism. Third, concerns over IP violations have emerged as a new driver of congressional trade criticism during the Trump and Biden administrations, reflecting the intensifying conflict in high-tech sectors among the two nations.

This article contributes to the literature of trade politics and firm lobbying in several ways. First, while traditional political-economy models of legislators' trade positions largely attribute firm influence to their material production profiles, I highlight how host-state policies and institutional environments shape firm interests (Lake 2009; Wellhausen 2015) and demonstrate a productive avenue to integrate insights from the deep trade agreement literature. Second, I identify the conditions under which GVC-engaged firms can fragment and mobilize anti-trade coalitions, contrary to the dominant pro-trade expectations of these firms emphasized in the literature (I. S. Kim 2017; Osgood 2018; Lee and Osgood 2021; Zhang 2023). Third, my empirical approach is among the first to systematically link firm lobbying and legislative behavior, opening wide-ranging applications for testing existing or new theories of business influence in both domestic and foreign policymaking.

2 Literature

2.1 Host state and foreign discrimination

In traditional trade politics models, politicians' positions on free trade reflect their constituents' interests, particularly those of businesses. Business preferences are structured by sectoral cleavages, most prominently the divide between export-oriented and import-competing industries (Baldwin and Magee 1998; Hiscox 2002). In the era of GVCs, however, the cleavages that shape trade politics increasingly cut across sectoral lines and have become far more mul-

tifaceted. As intra-firm trade and cross-border investment have become ubiquitous features of global trade, the once dominant export-versus-import divide has given way to new concerns, particularly input sourcing, foreign direct investment (FDI), and non-tariff barriers (Osgood 2018; Young 2016). At the same time, the politics of trade liberalization has moved beyond tariff reductions to encompass the negotiation of "deep" trade agreements. These agreements address a host of "behind-the-border" barriers to economic integration, including foreign investment policy, intellectual property rights, competition policy, government procurement, and many other regulatory issues (Baccini 2019). Deep trade agreements have been shown to increase both trade and FDI flows between member states (Dür, Baccini, and Elsig 2014; Büthe and Milner 2014).

In this context, firms' preferences have likewise become more multidimensional and heterogeneous than variation across sectors alone would suggest (Kim and Osgood 2019). Firms' support for free trade is shaped not only by their size, productivity, ability to product differentiate, but also by the degree of their participation in GVCs (Milner 1988; Osgood 2016; I. S. Kim 2017). Firms with deeper GVC participation, particularly through FDI or offshoring, are especially concerned with investment protection and dispute settlement (Kim et al. 2019) and tend to be key beneficiaries and strong proponents of deep, preferential trade agreements (Baccini, Pinto, and Weymouth 2017).

Building on this firm-centric literature, I argue that market access and production efficiency—two principal reasons for firms to engage in international trade and investment—are parameters over which host governments exercise control in ways that can disadvantage foreign firms in favor of domestic ones. Because firms' ability to secure market access and exploit production efficiency is contingent on host-government policies, their trade preferences are likewise shaped by those policies. Thus, I consider not only the material "production profiles" (Gourevitch 1986) of firms, but also endogenize their interests as "the product of political strategies and manipulations" (Lake 2009). In doing so, I highlight the host government's "space and autonomy to act against foreign firms' interests" (Wellhausen 2015). Such endogenization of firms' interests to host-government politics is especially warranted in the case of China, which practices a model of state capitalism characterized by strong state control over factors of pro-

duction (particularly credit allocation), the outsized presence of state-owned enterprises, and extensive intervention into strategic sectors of the economy (Tsai and Naughton 2015).

Foreign firms' access to the domestic market can be impeded by a range of policy barriers. More familiar instruments include the extensive use of licensing powers to regulate market entry and competition[3] (Pearson 2005) or prohibitions on foreign ownership (e.g., the 29 sectors listed in China's 2024 Negative List of Foreign Investment Access). A burgeoning literature has further examined less conventional instruments through which the Chinese state restricts market access and enacts protectionism, such as technology transfer requirements (Minnich, n.d.), politicized regulatory standards (Kim, Perlman, and Zeng 2025), and media bias (S. E. Kim 2018).

Apart from market access, foreign firms' ability to exploit local production efficiencies is also contingent on host-government policies. Firms' access—and the prices at which they can access—to a range of production inputs such as credit, land, energy, and intermediate goods are levers that host governments can control. For example, China has implemented a set of industrial policies to promote indigenous innovation and import substitution in strategic sectors since the late 2000s (Naughton 2021). The International Monetary Fund estimates that throughout the 2010s China devoted 4.4% of its GDP (around $783 billion) annually to industrial policy efforts with instruments such as cash subsidies, tax benefits, below-market-rate credit, and subsidized land, energy, or intermediate inputs (Garcia-Macia et al. 2025). Foreign firms without direct access to these subsidies or benefits are thus at a competitive disadvantage vis-à-vis other domestic firms.

Taking into account host-government policies and institutional environments reveals additional sources of variation in firms' experiences and preferences. On the one hand, sectoral variation still plays a role: industries deemed strategic to China's national security, such as telecommunications, are subject to more stringent regulations (Hsueh 2011). On the other hand, even within the same strategic sector, foreign firms' experiences of restrictiveness vary with their value-chain positions and bargaining power vis-à-vis Chinese counterparts (Min-

3. American businesses in China have consistently ranked "obtaining required licenses" and "inconsistent regulatory interpretation & unclear laws" as their top challenges in China throughout the 2010s, according to member surveys by American Chamber of Commerce in China."

nich, n.d.). Moreover, the multilevel and fragmented nature of the Chinese governance system generates substantial subnational variation in firms' experiences. Bureaucratic and political incentives have led subnational (provincial and local) governments to adopt more developmental or market-replacing policies, even when the central government implemented market-enhancing, regulatory reforms (Tan 2020). In the context of weaker formal institutions at the subnational levels, discretionary treatments by and particularistic ties with local officials can shape foreign firms' interests in China. For example, partnerships with state-owned enterprises have been shown to improve judicial outcomes for foreign firms (Chen and Xu 2023), while "special deals" with powerful local officials provide opportunities to break official rules or secure privileged access to resources (Bai, Hsieh, and Song 2020).

In short, in the contemporary era of complex GVCs, firms' preferences are deeply shaped by host-government policies over market access and production inputs. In China's state-capitalist context—where foreign restrictions, industrial policy, and subnational politics intersect—foreign firms face heterogeneous opportunities and constraints. These dynamics show that firm interests are endogenous to host-state political strategies and institutional environments.

2.2 Firm lobbying amid foreign discrimination

Firms facing foreign discriminatory policies that restrict market access have strong incentives to lobby policymakers to pressure foreign government, as illustrated by the political mobilization of American exporters against preferential trading zones established by the United Kingdom in the 1930s and by Europe in the 1960s (Dür 2007). Firms also play a crucial role in monitoring compliance with international laws and removing trade barriers by supplying information and subsidizing litigation efforts in World Trade Organization (WTO) disputes (Brutger 2024). This line of work parallels the lobbying literature in American politics, which has shifted from viewing lobbying as vote buying to emphasizing its informational utility in persuading potential allies (Schnakenberg 2017) and its role as a legislative subsidy for resource-constrained lawmakers (Hall and Deardorff 2006).

Firms disadvantaged by Chinese discriminatory policies hold unique, private information that is valuable in political persuasion, coalition building, and subsidizing legislative efforts.

Beyond these utilities, I argue that lobbying also serves to avoid political retaliation by the host state—an under-theorized dynamic in the trade politics literature. The Chinese government has imposed informal sanctions (Lim and Ferguson 2022) or mobilized consumer boycotts (Weiss et al. 2023) against foreign firms to pursue their own foreign policy interests. The US–China Business Council has cited similar concerns from its members that they "should have a reliable channel to report abuses and to appeal adverse decisions…without fear of retaliation" (USCBC 2017). Lobbying policymakers to speak on their behalf allows firms to raise concerns without publicly confronting Chinese authorities, which is especially important for firms with ongoing business or investment in China. Because lobbying records, though public, do not reveal direct connections between individual politicians and companies, this channel provides plausible deniability and allows firms hedge their political risks. (Indeed, one of the empirical contributions of this paper is to circumvent this challenge of opacity and estimate the effect of lobbying on legislators' positions.)

2.3 Legislative positions on free trade

Standard political-economy models assume that politicians respond to constituent interests to secure re-election. Accordingly, their positions on trade policy should reflect the voters' interests within their districts (Baldwin and Magee 1998; Beaulieu 2002). This voter-driven logic is especially visible in the "China Shock" literature. Studies show how Chinese import surges fueled protectionism, economic nationalism, or anti-China sentiment among politicians (Feigenbaum and Hall 2015; Kuk, Seligsohn, and Zhang 2018; Campello and Urdinez 2021; Meyerrose and Watson 2024). Feigenbaum and Hall (2015), for instance, argue that legislators respond to import shocks "in order to fend off electoral harm" because "*citizens demand* more protectionist policy" (emphasis added). While they find that China import shocks made Members of Congress vote in a more protectionist direction, they fail to find direct evidence that electoral responses are the main mechanism: incumbents in districts affected by import shocks are no more likely to face a primary challenge, receive fewer votes, or lose re-election.

Individual-level evidence for voter-driven models is also mixed. Some studies find that import competition heightens nationalism and authoritarian values among voters (Colantone

and Stanig 2018a, 2018b; Ballard-Rosa et al. 2021). Others show that trade is a low-salience issue among American voters, who rarely punish politicians for deviations (Guisinger 2009; Mansfield and Mutz 2009). Milner and Tingley (2011) highlight this tension: "Why would legislators form preferences around the economic effects of trade on their constituents if the constituents themselves do not?"

I propose one way to reconcile this tension: legislators may embrace anti-trade rhetoric that emphasizes foreign discrimination because such rhetoric triggers concerns about in-group favoritism, reciprocity, and fairness among voters—the main drivers of individual trade attitudes (Mutz and Kim 2017; Chilton, Milner, and Tingley 2020). By highlighting foreign discrimination and activating these sociotropic concerns, politicians can reduce the likelihood of electoral backlash even while adopting trade-skeptical or critical positions. Schweinberger (2022) provides direct survey evidence consistent with this argument: in the context of the first U.S.–China trade war, both American and Chinese respondents' support for cooperation fell more sharply when exposed to narratives about broken promises than when hearing narratives about positive behaviors from their counterpart.

3 Hypotheses

In line with the firm-centric literature on trade politics, I hypothesize how firms' engagement in GVCs with China shapes their trade preferences and, in turn, the positions of the legislators they lobby. Export orientation (Milner 1988; Kim and Osgood 2019) is a key predictor of firms' support for free trade, as only a small subset of large, productive firms can engage in exporting, and their revenues directly depend on the openness and stability of bilateral trade relations. But beyond simply exporting, input-sourcing and FDI have been shown to be the primary drivers of firms' support for free trade (Osgood 2018). I therefore expect that firms with different modes of GVC engagement will lobby legislators to promote free trade.

Hypothesis 1 *If firms export more to China, the legislators they lobby are less likely to criticize trade with China.*

Hypothesis 2 *If firms import more from China, the legislators they lobby are less likely to criticize trade with China.*

Hypothesis 3 *If firms have more FDI in China, the legislators they lobby are less likely to criticize trade with China.*

Building on my theory of foreign discrimination and business lobbying, I argue that firms facing discriminatory policies in China are more likely to lobby legislators to raise criticism of China's trade practices. Firms prefer lobbying over direct public criticism because lobbying provides deniability and reduces the risk of political retaliation. Legislators, for their part, are receptive to trade-critical rhetoric highlighting foreign discrimination because it activates sociotropic concerns among voters and mitigates the risk of electoral backlash.

Hypothesis 4 *If firms experience more discriminatory policies from China, the legislators they lobby are more likely to criticize trade with China.*

A key strength of my theoretical model is its dynamic nature: host governments' policies and institutional environments evolve in response to shifting developmental needs and strategic goals. Since the 2010s, China has promoted "indigenous innovation," culminating in the "Made in China 2025" initiative, which seeks to position Chinese firms at the commanding heights of high-tech industries. The plan explicitly calls for "promot[ing] a shift in the use of foreign capital from focusing on the introduction of technology, capital, and equipment to joint ventures and cooperative development" (PRC State Council 2015). This signals a more aggressive technology-absorption strategy, conditioning market access on mandated joint venture and technology transfers. I therefore expect that American firms increasingly encountered intellectual property and technology disputes with Chinese authorities and domestic firms beginning in the mid-2010s, and that they became more likely to lobby legislators to reflect these concerns.

Hypothesis 5 *If firms experience more intellectual property disputes with Chinese entities, the legislators they lobby are more likely to criticize free trade with China, especially beginning in the mid-2010s.*

4 Data and descriptive statistics

4.1 Measuring firms' interests in China

I propose a novel text-as-data approach to measure firms' interests in trading with China over time. The key assumption is that the intensity of a firm's particular type of interest is proportional to the frequency with which it is discussed in the company's financial documents. I compile financial filings of all publicly traded firms submitted to the U.S. Securities and Exchange Commission (SEC) as the main corpus. These filings contain rich textual information in mandated sections such as *Business Overview*, *Risk Factors*, and *Management's Discussion and Analysis* ("MD&A"), where public companies are required to accurately disclose their operations and discuss potential risks that might affect performance. The credibility of the information is reinforced by the fact that false or misleading statements to investors are subject to penalties under securities law.

Existing literature in finance and economics has long relied on SEC filings for quantitative textual analyses (Loughran and Mcdonald 2016). This approach is particularly well suited to measuring a range of China-related interests of U.S. firms. First, it provides comprehensive coverage of the universe of all U.S. public firms from the 1990s to the present. Second, because firms do not follow a consistent practice in reporting China-specific statistics (for example, while some large firms disclose revenues from China, many report only regional aggregates or provide no disaggregated data altogether), the current approach enables a standardized pipeline for constructing country-specific economic measures, such as firms' offshoring activities (Hoberg and Moon 2017). Lastly, it allows the construction of new measures that are not available in traditional economic data sources, such as the policy and geopolitical risks faced by firms (Hassan et al. 2025).

For the study period 2001–2022, I compile all annual reports (Form 10-K) filed with the SEC, using cleaned data from McDonald (2024).[4] I focus on reports that mention any China-related keywords (China, Chinese, Beijing, CCP, or PRC), resulting in a total of 58,960 filings submitted

4. I rely on McDonald (2024)'s "Stage One Parse" data which removed all the original markup language tags (HTML, XBRL, XML) from the raw text files. Procedures like removing stop words or stemming are not required because large language models perform better on the full context of natural language.

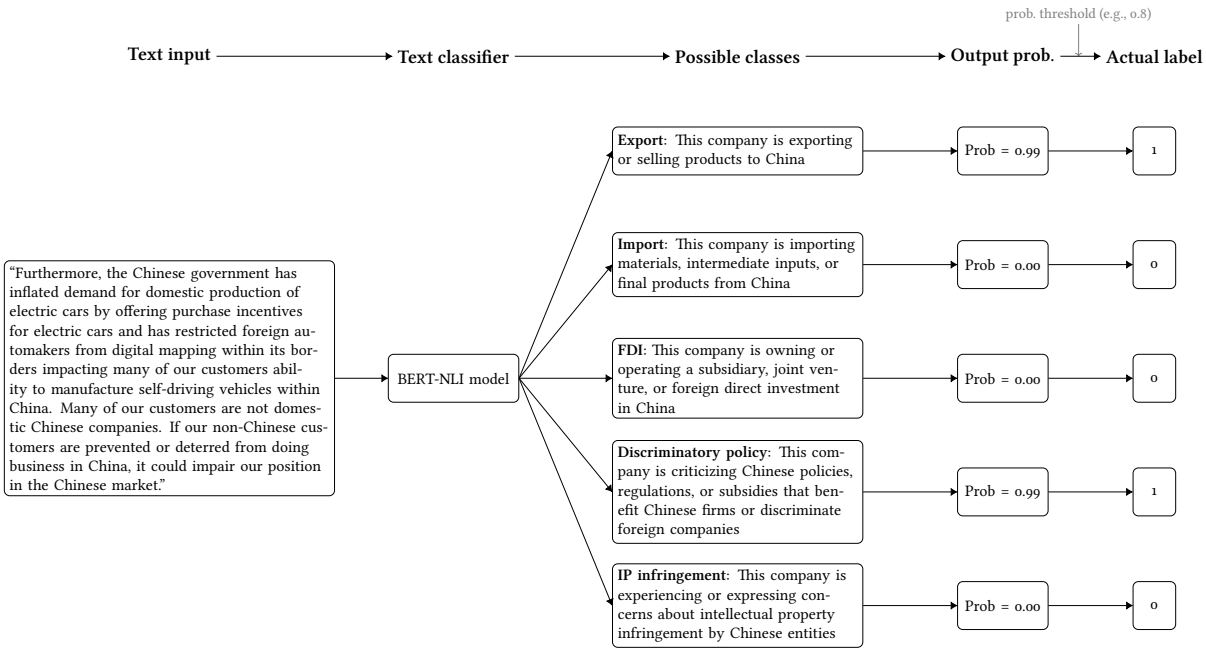

Figure 1: Annotation procedure applied to a sample text using a fine-tuned BERT-NLI classifier. The sample text is taken from the 2019 10-K filing by Veoneer, Inc., a Swedish-orign, Delaware-incorporated, Michigan-based company specializing in designing and manufacturing automotive safety electronics. In the sample text, Veoneer complained about Chinese policies that discriminated against foreign automakers and limited their ability to manufacture electric vehicles in China. These, in turn, hampered its sale of intermediate goods supplied to global automakers in China. The classifier appropriately assigns very high confidence scores to the export and discriminatory policy hypotheses and very low scores to the rest.

by 11,300 unique companies. Each filing is segmented into sentences, and I subset those that mention China. To enhance contextual understanding for downstream annotation, I augment each China-related sentence with the two preceding and following sentences. This procedure yields 1,063,126 China-relevant sentences, with a median word count of 149. These sentences constitute the input for the annotation procedure that generates China-specific labels at the firm-year level.

To efficiently annotate this large corpus of filings and generate meaningful labels for downstream analyses, I employ a class of BERT-NLI classifiers[5] that are increasingly adopted in political science for text annotation (Laurer et al. 2024; Burnham 2024). These open-source,

5. BERT-NLI refers to a class of transformer-based models pre-trained for Natural Language Inference (NLI). In NLI, the task is to determine whether a human reading of the text T would infer that the hypothesis H is most likely true; in other words, whether T *entails* H (Dagan, Glickman, and Magnini 2006). Text annotation can be formulated as a NLI task. For example, given the text "Too many American jobs were outsourced to China," the model would likely infer that the hypothesis "The author of this text supports *less* trade with China" is true. BERT-NLI models output confidence scores, which are logits scores transformed via the softmax function to approximate probabilities.

encoder-only models offer performance comparable to their proprietary, decoder-only counterparts in classification tasks, such as chatGPT. Most importantly, they also mitigate concerns for reproducibility and transparency that plague LLM-powered research (Spirling 2023), since the model parameters can be inspected, the inference procedure directly examined, and the results reliably replicated. Moreover, given the large volume of text, these models substantively reduce inference costs by orders of magnitude compared to proprietary alternatives, which also enable efficient replication of model results.

Figure 1 illustrates the annotation pipeline. A China-relevant sentence is input into the BERT-NLI classifier and evaluated against five non-mutually exclusive classes of firm interests, formulated as hypotheses: export, import, FDI, discriminatory policy, and IP infringement. The classifier outputs confidence scores for each class, reflecting its confidence that the corresponding hypothesis is entailed. Binary values (0 or 1) are then assigned to the class by applying a probability threshold, such as 0.8.[6] By iterating this procedure over all China-related sentences, I construct firm–year-level measures of the frequency with firms mention specific concerns in their filings.

Throughout this study, I employ `deberta-v3-large-zeroshot-v2.0` as the baseline model, which is an open-source, 435-million-parameter model trained on a broad set of NLI datasets (Laurer et al. 2024). To further improve and evaluate model performance, I manually annotated 400 sentences randomly sampled across five classes of firms' interests, generating 2,000 human labels to fine-tune the baseline model. The fine-tuned classifier achieves a macro-average F1 of 0.87 and a balanced accuracy of 0.84, which exceed the performance typically reported in recent benchmark studies (Laurer et al. 2024). The technical details of model fine-tuning, inference, and evaluation are reported in the Appendix.

Figure 2 presents the frequency with which firms discussed five types of interests related to china in their annual filings from 2001 to 2022. On average, the most frequently discussed interest concerns FDI. Mentions of FDI increased after China's accession to the WTO in 2001,

6. In practice, researchers usually use the `argmax` function, which selects the label with the highest probability. In the context of two labels (entailment versus non-entailment), this corresponds to a de facto threshold of 0.5. In this study, I adopt a more conservative threshold of 0.8 as the baseline to reduce false positives. Because firms have multiple opportunities to discuss a particular type of interest throughout their filings, the potential cost of allowing some false negatives is tolerable. In robustness check exercises, I use alternative thresholds (0.7 and 0.9) to show that the main results are not driven by this choice.

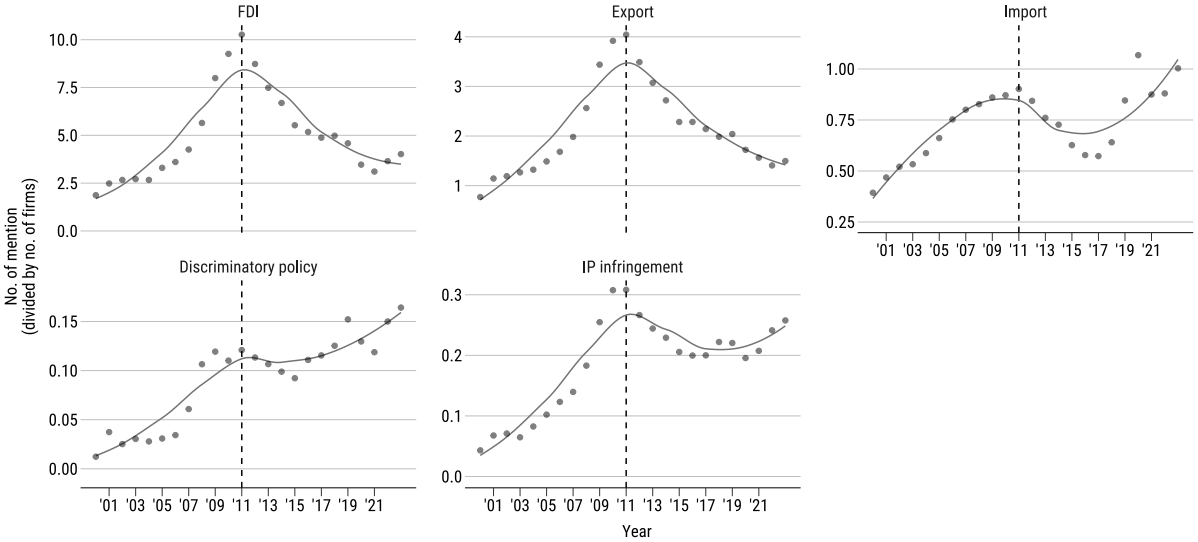

Figure 2: Frequency of firm-level discussions of business interests in China, normalized by the annual number of firms mentioning China in their filings.

rising from an average of 3.5 per firm during 2001–2008 to a peak of 9.2 during 2009–2011. After 2011, however, the frequency declined to an average of 5.1. A similar pattern is observed for export interests, which also peaked in 2011 before declining.[7] Import interests were mentioned less frequently than FDI and export, but unlike the other two, discussions of imports rebounded in 2018 after the post-2011 decline. It was likely driven by a combination of Trump's first trade war with China in 2018, the disruption to global supply chains by COVID since 2020, and the Biden administration's continuation of tariff and trade-restriction policies, all of which prompted firms to highlight their reliance on Chinese imports and the risks associated with supply disruptions.

Beyond discussions of firms' GVC engagement with China, their complaints about Chinese discriminatory policies increased substantively after 2006, quadrupling from an average of 0.03 mentions per firm during 2001-2007 to 0.12 from 2008 onward. Discussions on IP infringement show even earlier signs of intensification: their frequency rose steadily in the

7. While multiple factors may explain the inflection point around 2011 and further research is warranted, these patterns are consistent with China's efforts to promote domestic consumption and promote industry upgrading, as outlined in the 12th Five-Year Plan (2011-2015). This policy pivot was accelerated by the 2008 financial crisis, which exposed the vulnerabilities of the export-driven growth model led by foreign demand. Evidence from global input-output data also suggests that China's reliance on imported manufacturing intermediates peaked in 2005 and has declined sharply since then (Baldwin, Freeman, and Theodorakopoulos 2023). By the 2010s, China has become the least dependent on foreign suppliers among the world's four largest manufacturing economies (the U.S., Germany, Japan, and China), indicating some success in import substitution.

mid-2000s, peaked in the early 2010s at an average of 0.30 mentions per firm, and remained elevated throughout the 2010s at above 0.20.

To my knowledge, this is the first systematic dataset documenting U.S. firms' diverse interests in China, encompassing not only traditional concepts such as exports, imports, and FDI but also less conventional ones such as discriminatory policy and IP infringement. These measures reveal rich temporal variation over the past two decades and underscore the value of my text-as-data approach for measuring firm interests.

4.2 Measuring legislators' anti-trade positions

To measure the policy positions of members of Congress on free trade with China, I adopt a text-as-data approach based on legislative speech. The trade politics literature has traditionally relied on roll-call votes to measure legislators' positions on free trade. However, roll-call data often exhibit low variation as Members of Congress often vote strategically along party lines or in an unanimous fashion. Schwarz, Traber, and Benoit (2017) show that text-based scaling of ideal points using parliamentary speeches reveals greater intra-party variation than those estimated from roll call data. This problem is compounded by the fact that very few bills seeking to overhaul U.S.–China trade relations have reached a floor vote since 2000, when Congress granted China Permanent Normal Trade Relations (PNTR).[8] Legislative speech, by contrast, is a core form of political position-taking in its own right, particularly responsive to major political events and revealing of political disagreements even when no votes occur, as in debates over the U.S.–China trade wars (Lauderdale and Herzog 2016).

I operationalize legislators' positions on free trade with China in two ways. First, I construct a broad measure: the number of speeches that express opposition to free trade with China. However, since legislators may oppose free trade on grounds unrelated to business, such as national security, I further construct two issue-specific measures: the number of speeches criticizing China for (a) "engaging in unfair trade practices" and (b) "stealing American technology or intellectual property." I choose two grievances since they have been among

8. One notable exception is the Uyghur Forced Labor Prevention Act passed in 2021, which establishes a rebuttable presumption that any import from Xinjiang, China is tinted with forced labor and therefore barred from entry into the US. Since the bill received overwhelming bipartisan support, the voting record reveals virtually no variation: it was passed in the House 428-1 and unanimously passed the Senate during the 117th Congress.

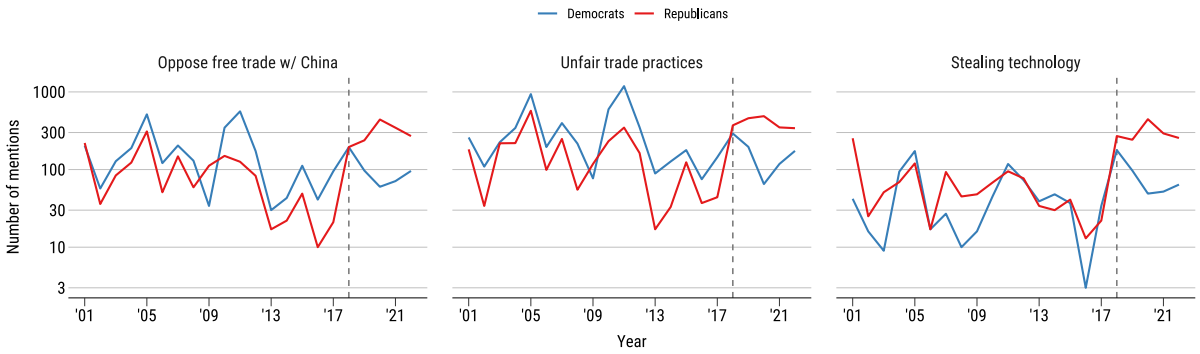

Figure 3: Members of Congress' criticisms of trade with China, by party, 2001 to 2022 (107th to 117th Congress).

the most prominent concerns of U.S. policymakers critical of trade with China (Menaldo and Wittstock 2021). In addition, these two grievances map respectively onto firms' concerns with China's discriminatory policies and intellectual property infringement, aligning more closely with my theoretical arguments of firm lobbying shaping legislative behaviors.

I employ a text annotation pipeline, similar to the one used for firms' financial filings, on the full corpus of the Congressional Record. I collect the full Congressional Record from 2001 to 2022 using the `congressional-record` parser (Judd et al. 2017). I segment each speech into sentences, subset those that mentioned China, and augment them with two preceding and following sentences for additional context. This procedure yields 72,419 China-related sentences from 1,178 unique legislators, with a median sentence length of 113 words. I use the same baseline BERT-NLI model to annotate the China-relevant speeches with three non-mutually exclusive classes: opposing free trade with China, criticizing China's unfair trade practices, and criticizing China for stealing American technology or IP.[9]

Figure 3 shows the frequency of criticisms of trade with China by Members of Congress over time. Criticisms of free trade with China in general, and of China's unfair trade practices in particular, peaked in 2005 and again during 2010-2012. These surges likely correspond with episodes of congressional pressure over the issue of currency manipulation.[10] Throughout the

9. Evaluation of model performance based on 200 randomly sampled, manually coded speeches achieves a balanced accuracy of 0.874. Given the satisfactory performance without fine-tuning, I use the baseline `deberta-v3-large-zeroshot-v2.0` model.

10. During the 109th Congress (2005-2006), for example, Senators Charles Schumer (D-NY) and Lindsey Graham (R-SC) introduced a bill (S.295) that called for a 27.5% tariff on Chinese goods if China failed to address RMB undervaluation. Although the Schumer-Graham bill eventually died in the Senate Finance Committee, it exerted significant pressure on the Bush administration in shaping the bilateral dialogue. A similar pattern

2000s, the Chinese currency was undervalued by approximately 25% (Rodrik 2010) through central bank intervention in currency markets, which led U.S. policymakers to accuse China of manipulating its currency to boost export competitiveness at the expense of U.S. exporters.

Democrats were consistently more critical of free trade with China and more vocal about China's trade practices than Republicans throughout the 2000s and early 2010s. However, this pattern reversed sharply after 2018, when the Trump administration launched the trade war with China, with Republicans becoming the dominant critics of trade—a striking partisan realignment on the issue of free trade that persisted even in the first two years of the Biden era. Meanwhile, accusations of China's technology theft were voiced by members of both parties throughout the 2000s. The issue has regained prominence since the first Trump administration, likely because the United States Trade Representative launched a "Section 301" investigation on China's policies and practices related to technology transfer and IP in August 2017 and concluded in early 2018 that they were "unreasonable or discriminatory" to U.S. commerce (USTR 2018). In fact, these findings provided the formal justification for the Trump administration's first tariff actions against China in 2018.

The speech-based measures of legislators' positions on free trade with China correspond with major episodes of heightened tensions in U.S.–China trade relations and reveal interesting temporal and intra-party variation. I create legislator-year-level total numbers of speeches for three types of trade criticisms respectively, which serve as the outcome variables in the main analyses.

4.3 Linking firm concerns with legislative rhetoric

The remaining empirical challenge is to develop a framework that links firm concerns to legislative behaviors and enables regression analyses at the politician-year level—an approach not yet established in the literature. I propose a novel measure of firms' *lobby-transmitted interests* whereby the concerns of firms are relayed to legislators through their lobbying connections. I

of congressional pressure on the executive emerged during the Obama administration, particularly around the annual U.S.–China Strategic and Economic Dialogue. In 2010, 130 members of Congress signed a joint letter, urging the Commerce and Treasury Departments to address China's currency manipulation; the following year, the Senate passed the "Currency Exchange Rate Oversight Reform Act," which drew strong condemnation from China. By the 2013 Dialogue, the Obama administration noted that "the RMB exchange rate has appreciated" and that "significant advances" had been made over the preceding four years of engagement (Burns 2013).

leverage the firm-legislator network data from the LobbyView dataset (I. S. Kim 2018). In the lobbying networks, a tie between a firm (or any interest group) and a member of Congress is established when the former lobbied on a bill sponsored by the latter in a given year. This operationalization provides only an indirect measure of lobbying connections, given the limitation that lobbying disclosure documents do not identify the specific legislators targeted by a firm's lobbying activities. To guard against erroneously inferring ties, I impose a threshold requiring a firm lobby on at least five bills sponsored by the legislator in a given year to qualify as having a lobbying connection.[11] As Kim and Kunisky (2020) argue, "*recurring* instances of lobbying that involve the same interest group and sponsor on numerous bills do reliably indicate a shared involvement on specific political issues."

I compute novel measures of lobby-transmitted interests through matrix multiplication for firm f with specific interest i lobbying legislator l at time t:

$$x_{l,i,t} = \sum_{f=1}^{F} \sqrt{w_{l,f,t} \cdot s_{f,i,t}} \tag{1}$$

The first matrix $w_{l,f,t}$ comes from the firm-legislator network data, representing the weight of the lobbying tie between legislator l and firm f at time t. The second matrix $s_{f,i,t}$ comes from firms' interests in China measured from the financial filing annotation, capturing the strength of firm f's interest in issue i (e.g., the number of mentions of export interest) at time t. Their product $x_{l,i,t}$, therefore, yields the weighted sum of a specific firm interest to which a legislator is exposed, aggregated across all firms with which they have lobbying ties. The square-root transformation is applied to both the lobbying tie weights and the strengths of firm interests to capture diminishing returns across both dimensions, reduce the influence of extreme values, and conveniently accommodate zeros. More intuitively, the lobby-transmitted interests capture legislators' exposure to specific firm interests, with greater influence coming from interests that firms discuss more frequently, from firms with stronger lobbying ties, and from more firms reiterating the same concern.

After harmonizing multiple types of company identifiers, I identify 2,051,034 lobbying ties

11. I set the threshold at 5, one above the median number of ties (4) in the lobbying network data. In robustness checks, I use a higher threshold of 10 ties (the upper quartile value is 9) and show that the main results are not sensitive to the choice of threshold.

that can be matched to firms appearing in my text annotation pipeline and to specific legislators over the years. I focus on 968,079 of them which involve at least 5 bills between the firm-legislator dyads. Based on the operations described above, I construct five type of firms' lobby-transmitted interests at the politician-year level: export, import, FDI, discriminatory policy, and IP infringement. These lobby-transmitted interests serve as the main explanatory variables in the regression analyses.

5 Empirical strategy

To estimate the relationships between firms' lobby-transmitted interests and legislators' rhetoric on trade with China, I fit a series of negative binomial models for count data with the following setup:[12]

$$\ln(\mathbb{E}[Y_{it}]) = \alpha Y_{it-1} + \beta X_{it} + \theta Z_{it} + \gamma_s + \delta_t + \epsilon_{it} \tag{2}$$

The dependent variable Y_{it} is the number of instances in which legislator i in year t expressed opposition to free trade with China (or criticized China for unfair trade practices, or technology theft, as alternative measures). The lagged dependent variable Y_{it-1} is included on the right-hand side to account for serial correlation in legislative rhetoric. The key explanatory variables are X_{it}, which measure legislators' exposure to firms' lobby-transmitted interests.

To address endogeneity and to strengthen a causal interpretation of the regression estimates from observational data, I employ the following strategies. First, I incorporate state fixed effects (γ_s) and year fixed effects (δ_t) in all models to absorb state-specific, time invariant unobservables and year-specific shocks. Second, I include a set of legislator-year-level control variables Z_{it}: party affiliation and left-right ideology (the first dimension of the DW-NOMINATE scores) to account for partisan and ideological differences in attracting business lobbying and in expressing trade criticisms; chamber (House versus Senate) to account for differences in responsiveness to societal interests; tenure (measured in years) to account for dif-

12. The outcome variables–the number of speeches in which Members of Congress criticized trade with China–contain many zeros, as 90% of the politician-year observations involve no China-related discussion. Transformation of outcome Y using $log(Y + c)$ or the inverse hyperbolic sine (IHS) function is known to make coefficients sensitive to the arbitrary choice of parameters and the unit of measurement (Mullahy and Norton 2024).

ferences in seniority and opportunities for legislative activities; membership in finance-related committees and foreign-affairs-related committees, as the former are traditionally more pro-trade while the latter are more critical of trade for non-economic reasons.[13]

Third, I rule out several alternative explanations that may confound the relationships between firms' lobby-transmitted interests and legislative rhetoric: (1) district-level sectoral and economic characteristics, (2) districts' exposure to China's import shock, (3) politicians' military hawkishness, and (4) contemporaneous bias.

District-level economic characteristics, particularly sectoral employment, have been a dominant explanation of legislative behaviors in the trade politics literature (Baldwin and Magee 1998), and they can also influence patterns of business lobbying. For all House legislator-year observations, I compile and control for total employment in nine major sectors at the congressional-district (CD) level,[14] derived from the U.S. Census' County Business Patterns (CBP) data using a population-weighted apportionment (Autor et al. 2020; Ferrara, Testa, and Zhou 2024).[15] Additionally, I include the district-level percentage of bachelor's degree holders to account for differences in human capital and education, as well as the unemployment rate as further controls.[16]

In the same vein, the China shock literature offers a similar prediction: districts with sectoral characteristics most exposed to China's import competition are more likely to see rising

13. I consider the following as "finance-related committees": Ways and Means Committee and Financial Services Committee on the House side; and Finance Committee and Banking, Housing, and Urban Affairs Committee on the Senate side. "Foreign-affairs-related committees" include the House and Senate Foreign Affairs, Armed Services, and Intelligence Committees. Committee membership data are from Stewart (2021a, 2021b).

14. The nine major sectors include Natural Resources and Mining (NASIC 2-digit code 11, 21); Construction (23); Manufacturing (31-33); Trade, Transportation, and Utilities (42, 44-45, 48-49; 22); Information (51); Financial Activities (52, 53); Professional and Business Services (54, 55, 56); Education and Health Services (61, 62); and Leisure and Hospitality (71, 72).

15. Matching county-level data (e.g., County Business Patterns from the U.S. Census) to congressional districts is a non-trivial task. The first challenge is that congressional districts are redrawn every ten years following the decennial Census, with occasional redistricting during the intermediate period in some states. The second challenge is that congressional districts are not coterminous with county boundaries, spanning or cutting across multiple counties. The most significant issue, however, is that CBP data reflect the number of employees working at establishments located in a given county, not the number of people living in that county employed in specific sectors. To address the first two issue, Ferrara, Testa, and Zhou (2024) develop crosswalk tables that map counties to congressional districts over time, using weights based on the share of a county's population residing within each district. But apportioning county-level CBP data to congressional districts based on population weights assumes that the spatial distribution of establishments is the same as that of the population—an imperfect assumption which nevertheless provides a reasonable and internally consistent proxy for the sectoral characteristics at the congressional district level.

16. County-level data in education and unemployment are from Economic Research Service of the U.S. Department of Agriculture. They are apportioned to the CD level manually as described above.

protectionism among their voters and representatives (Feigenbaum and Hall 2015). Although most studies focus on the role of voters, businesses are also constituents adversely impacted by import surges and may increase their lobbying. To account for this potential confounder, I extend the import exposure to China per U.S. worker data from Autor et al. (2020) to cover my study period using the population-weighted apportionment described above.

Apart from economic explanations of legislative behavior, I also account for legislators' military hawkishness toward China. Firms may target and lobby hawkish legislators to apply political pressure on China, but these legislators may oppose trade with China on national security grounds rather than business concerns. I measure hawkishness by the number of times legislators described China as a military threat to the U.S. in a given year, based the Congressional Record and using the same text annotation pipeline described in the measurement section. Lastly, I consider a potential source of endogeneity arising from contemporaneous bias. In a given year, legislators' speeches may precede firms' lobbying activities, thereby reversing the causal direction.[17] To address this concern, I include a one-year lag of firms' lobby-transmitted interests as the main predictors in additional models.

To address the remaining issue of confoundedness, I conduct a sensitivity analysis to evaluate the robustness of the main results and to benchmark the minimum strength of unobserved confounders required to overturn the conclusion (Cinelli and Hazlett 2020). Summary statistics of all the variables in the analyses are presented in Table A.1 in the Appendix. For all regression models, I cluster standard errors at the member-of-Congress level to account for within-member correlation in the error terms.

17. Lobbying activities are reported on a quarterly basis (no later than 20 days after the end of each quarter) mandated by the Lobbying Disclosure Act of 1995. So, the timing of the reported lobbying activity generally coincides with the reporting quarter, though it may still be preceded by legislators' speeches. By contrasts, Form 10-K filings submitted to the SEC in year t typically cover the previous fiscal year $t - 1$, so there is a more substantial time lag between firms raising certain concerns and legislators speaking on the issues.

6 Results

6.1 Explaining legislators' trade-critical rhetoric

Regression results broadly support that legislators lobbied by firms which referenced discriminatory policies in China are more likely to oppose free trade with China and criticize China's unfair trade practices. By contrasts, firms with GVC interests in China, particularly in import, tend to moderate trade-critical rhetoric. I rule out several alternative explanations, including district-level economic characteristics, Chinese import shocks, legislators' military hawkishness, or contemporaneous bias. They are also robust to a host of alternative specifications and ways of constructing lobbying ties.

The regression estimates of legislators' trade rhetoric on firms' lobby-transmitted interests are reported in Table 1. Model 1 shows the baseline results of four categories of lobby-transmitted interests. Firms' concern about discriminatory policies predicts greater legislators' opposition to free trade with China, while firms' concerns about imports tend to moderate such anti-trade rhetoric. The coefficients of export and FDI interests, however, are close to zero and statistically insignificant.

Model 2 includes legislator-level controls and a lagged dependent variable, showing that the effects of firms' interests persist. It also allows for comparisons of their effect sizes with other variables. For example, a two-standard-deviation increase in import concerns is associated with 87.0% less rhetoric opposing free trade with China, whereas being a Republican, traditionally more pro-trade, is associated only with 59.5% less anti-trade rhetoric when compared to a Democrat.[18] By contrast, a two-standard-deviation increase in discriminatory policies concerns is associated with 209.5% more anti-trade rhetoric, which is also larger than the effect of being a Senator (159.3% increase relative to a House representative) or a Democratic (147.1% increase relative to a Republican). Perhaps surprisingly, the positive coefficient on left-right ideology indicates that more economically conservative legislators are *more* likely to criticize free trade, with a two–standard-deviation increase in conservatism associated with 191.5% more anti-trade rhetoric.

18. Following Gelman (2008), a change from 0 to 1 in a binary variable is comparable in magnitude to a two-standard-deviation change (mean $\pm$ 1 standard deviation) in a continuous variable.

Table 1: Firms' lobby-transmitted interests and legislators' rhetoric on trade with China, 2001 to 2022.

	Oppose trade		Unfair trade	Oppose trade
	(1)	(2)	(3)	(4)
Export	-0.002	0.003	0.0004	-0.0005
	(0.004)	(0.003)	(0.004)	(0.003)
Import	-0.037***	-0.029***	-0.027***	-0.015***
	(0.005)	(0.004)	(0.004)	(0.003)
FDI	0.001	-0.0003	0.001	0.002
	(0.002)	(0.002)	(0.002)	(0.002)
Discriminatory policies	0.057***	0.054***	0.060***	0.011
	(0.009)	(0.008)	(0.008)	(0.008)
Party: Independent		2.59*	2.32*	0.476
		(1.10)	(0.992)	(0.561)
Party: Republican		-0.905*	-0.877*	-0.254
		(0.387)	(0.371)	(0.303)
Chamber: Senate		0.953***	1.23***	0.250*
		(0.150)	(0.143)	(0.122)
Left-right ideology		1.23*	1.37**	0.394
		(0.486)	(0.470)	(0.350)
Finance committees		-0.101	0.146	-0.236**
		(0.105)	(0.104)	(0.091)
Foreign affairs committees		0.308**	0.171	0.329***
		(0.107)	(0.112)	(0.084)
Tenure (years)		0.041***	0.040***	0.024***
		(0.007)	(0.006)	(0.005)
Unfair trade				0.306***
				(0.038)
Observations	12,040	11,198	11,198	11,198
Lagged DV_{t-1}	✗	✓	✓	✓
State fixed effects (50)	✓	✓	✓	✓
Year fixed effects (22)	✓	✓	✓	✓

Notes. Coefficients from negative binomial models. Standard errors clustered by member ID are in parentheses. $^{+}p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

This seemingly paradoxical result can be explained by further unpacking the framing of anti-trade rhetoric. Model 3 addresses this by using criticisms of China's "unfair trade" practices as the dependent variable. The results closely mirror those in Model 2, suggesting that much of legislators' opposition to free trade with China is rooted in concerns about unfair trade practices. Within both parties, including Republicans, more economically conservative members are more likely voice such concerns. This finding echoes Nollen and Quinn (1994), who argue that "fair trade" is a distinctive issue, rather than merely "protectionism in disguise," and garners support even from legislators with strong financial backing from international businesses.

This interpretation is supported by Model 4, which uses opposition to free trade as the dependent variable but includes legislators' unfair trade criticism as a control, thereby isolating anti-trade rhetoric not explained by unfair trade concerns. Once such concerns are partialled out, the discriminatory policies variable no longer predicts anti-trade rhetoric, suggesting that firms' lobbying is linked to legislators' specific criticism of China's trade practices rather than to generic anti-trade sentiments. Similarly, economic conservatism also ceases to be significantly associated with anti-trade rhetoric. By contrast, import concerns continue to predict less anti-trade rhetoric, suggesting that their moderating effect operates broadly across different framings of trade criticism.

6.2 Accounting for alternative explanations

Next, I present evidence that the relationships between firms' lobby-transmitted interests and legislators' anti-trade rhetoric are robust to several alternative explanations. Figure 4 visualizes the effect sizes of firm interests in comparison with other key variables, while the full regression results are present in Table A.2 in the Appendix.

Differences in district-level employment across sectors and in levels of human capital and unemployment—the dominant explanations in the trade politics literature—do not account for the relationships between firm lobbying and legislative rhetoric, as shown in the top panel of Figure 4. Most sectoral employment and economic indicators fail to reach conventional levels of statistical significance, with only a few exceptions. Higher employment in finance and

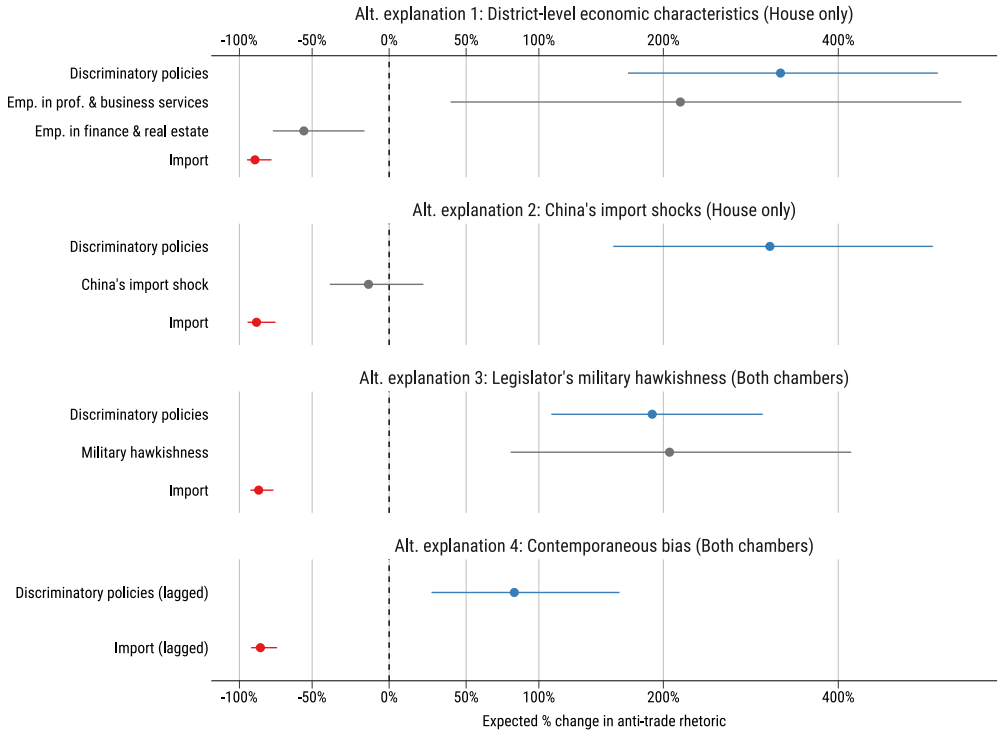

Figure 4: Accounting for alternative explanations of firm lobbying and anti-trade rhetoric. The expected percentage change in rhetoric is computed from a hypothetical two-standard-deviation change in the variable of interest.

real estate within a congressional district significantly predicts less anti-trade rhetoric from its House representative, while employment in professional and business services predicts more of it. This pattern reveals a notable fragmentation within the broader service sector: finance-related industries (including insurance) tend to favor free trade with China, while other professional and business services are more critical of the trade regime. This fragmentation contrasts with the unity of "pro-trade" service sector often portrayed in the literature (Baccini, Osgood, and Weymouth 2019) but is consistent with a theory of differential market access, where the Chinese state selectively opens sectors and grants access to certain firms.

China's import shock does not explain the relationships between firm lobbying and legislators' anti-trade rhetoric, as shown in the second panel. In fact, it does not exhibit any significant association with trade rhetoric even when firm interests are excluded from the model, at least during the study period of 2001–2022. This is surprising given the centrality on the China shock in recent trade politics scholarship (Feigenbaum and Hall 2015; Kuk, Seligsohn, and Zhang 2018). It also call into questions the voter-driven mechanism of legis-

lators' trade positions that underpins much of this literature in the U.S. context (Guisinger 2009). To further assess whether electoral politics intersects with firm lobbying, I include the interactions between firms interests and the electoral competitiveness of the district. The results, reported in Table A.3 in the Appendix, indicate that the effects of firm lobbying do not significantly vary with respect to district competitiveness.[19] These findings suggest that firm lobbying constitutes an independent mechanism in shaping legislators' anti-trade rhetoric on China, distinct from China's import surges or the voter-driven mechanism underpinning the China shock studies.

Apart from district-level economic characteristics, legislators' military hawkishness also does not confound relationships between firm lobbying and legislators' anti-trade rhetoric, as shown in the third panel. While legislators who characterize China as a military threat to the U.S. do criticize free trade with China more frequently, its effect size is comparable to that of firms' concerns about discriminatory policies and does not render the latter spurious. Lastly, I address the potential contemporaneous bias by using the lagged firms' lobby-transmitted concerns in the model. The effect size of firms' concerns about discriminatory policies shrinks approximately by half but remains statistically significant, whereas that of import interests remain largely unchanged. The evidence suggests that contemporaneous bias may be present but does not invalidate the observed associations.

To address the remaining concern of omitted variable bias, I conduct a sensitivity analysis to evaluate the robustness of the main results (see Figure A.1 in the Appendix). I benchmark the minimum strength of potential unobserved confounders required to invalidate the associations, using the effect of partisanship (being a Republican compared to a Democrat) as reference. Under the maximally conservative scenario where unobserved confounders explain *all* residual variance in legislative rhetoric, such confounders would need to be nearly twice as strongly associated with firms' lobby-transmitted interests as partisanship is, in order to fully explain away the observed effects. Given that the previous analyses have accounted for a range of legislator-level characteristics and alternative explanations encompassing economic,

19. The interaction term between firms' discriminatory policy concerns and district competitiveness is marginally significant at the 0.10 level, but this significance does not hold under alternative clustering of standard errors, such as clustering at both the member and state levels.

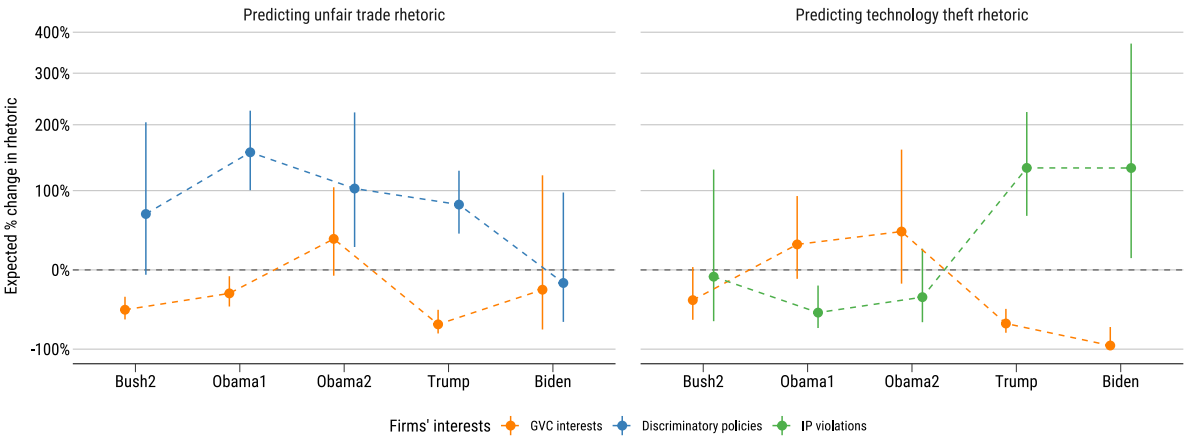

Figure 5: Temporal dynamics of firm lobbying and legislative rhetoric on trade practices and technology theft across presidencies. The expected percentage change in rhetoric is computed from a hypothetical one-standard-deviation change in the variable of interest.

ideological, and temporal confounders, the sensitivity analysis indicates that unobserved confounding is unlikely to pose a serious threat to a causal interpretation of the main findings.

6.3 Temporal variation and the resurgence of IP conflicts

I further exploit the rich temporal variation in my data to examine to heterogeneous effects of firm lobbying on legislative rhetoric. I focus on two specific framings of legislators' criticisms of China as separate outcomes: unfair trade practices and technology theft. To facilitate temporal comparison, I construct a new variable, GVC interests, which is the sum of export, import, and FDI interests. For criticisms of unfair trade practices, I interact firms' GVC interests and discriminatory policy concerns respectively with indicators for the presidential term and employ state and presidency fixed effects. For criticisms of technology theft, the specification is the same except that the main explanatory variable is firms' concerns about IP infringement. The full results are reported in Tables A.4 and A.5 in the Appendix.

Figure 5 visualizes the temporal dynamics of firm lobbying and legislators' criticisms of China. The left panel shows that firms' complaints about China's discriminatory policies were a key driver of legislative rhetoric on unfair trade practices since the second Bush administration. The effect peaked during the first Obama administration and had since gradually subsided, disappearing by the Biden administration. Meanwhile, the moderating effects of GVC interests on unfair trade rhetoric have been inconsistent: they dampened such rhetoric

during the second Bush, first Obama, and Trump administrations but failed to do so in the second Obama and Biden administrations. Further disaggregating GVC interests into the original categories reveals that FDI interests actually predict *more* unfair trade rhetoric during those administrations (Model 2 of Table A.4), driving the occasional non-significance and sign reversal observed in the left panel.

While firm lobbying ceased to drive unfair trade criticism in the Biden era, firms' concerns about IP infringement have fueled a separate rhetoric accusing China of technology theft in recent administrations. As shown in the right panel, firms' IP-related concerns did not predict more legislative rhetoric about technology theft from the second Bush through the second Obama administrations (indeed, the coefficient was even negative during the first Obama term). However, these concerns became a positive and significant predictor during the Trump and Biden administrations. On the other hand, the moderating effects of GVC interests again exhibit inconsistency across administrations, with a similar reversal of sign observed during the Obama terms driven by FDI interests.

These results reveal the rich temporal dynamics of the relationships between firm lobbying and legislative rhetoric. Not only do firm interests in China change, their political salience and ability to shape legislative rhetoric and specific framings also shift. Such changes arise from both the supply and demand sides of legislative rhetoric. The finding that firms' IP-related concerns fueled legislative rhetoric about technology theft only during the Trump and Biden administrations can partly be explained by the agenda-setting and coalition-building power of the president in foreign policy (Milner and Tingley 2011). For example, the Trump administration launched the first trade war, justifying it with the USTR's Section 301 investigation into China's technology transfer and IP practices, while the Biden administration continued to treat competition with China over advanced technology sectors as a foreign policy priority. The agenda set by these presidents created political opportunities for legislators to adopt more critical rhetoric toward trade with China, particularly on its IP practices.

However, this explanation is incomplete without accounting for the demand side of trade-critical rhetoric. I show that firms' concerns about IP infringement in China are a key driver behind such rhetoric. Part of this shift in firms' experiences can be attributed to changes in

China's industrial policy and developmental priorities. Beginning around 2010, China implemented a coherent set of industrial policies to promote indigenous firms in "strategic emerging industries" (Naughton 2021), culminating in the "Made in China 2025" plan announced in 2015. China instituted a set of technology absorption polices, including conditioning foreign market access on technology transfer to domestic firms (Minnich, n.d.). Consistent with my theory emphasizing the host government's capacity to enact discriminatory policies to gate market access and production resources, foreign firms might demand trade-critical rhetoric from legislators as a means of exerting pressure and influencing China's policy choices.

This interpretation is also consistent with the surprising sign reversal of GVC interests, which became positively associated with trade-critical rhetoric during the second Obama administration. The fact that firms with FDI interests in China lobbied legislators to criticize China's trade and IP practices runs counter to the pro-trade expectations in the literature. It may instead point to a more fundamental policy and institutional shift in how China treats foreign firms. The causes of this changing attitude from firms with FDI in China warrant further research.

6.4 Robustness

I assess the robustness of the main results (Model 2 of Table 1) using a series of alternative model specifications and variable constructions. First, I employ a OLS model instead of a negative binomial model (Table A.6 in the Appendix). Separately, I employ a two-way fixed effects specification at the legislator (instead of state) and year levels, with the trade-off of removing all time-invariant controls such as party, chamber, committee assignments, and ideology (Table A.7). Both analyses show that firms' concerns about discriminatory policies and import retain their expected signs and statistical significance.

Throughout the empirical section, all analyses rely on text-annotation generated variables: firms' concerns derived on financial filings and legislative rhetoric from the Congressional Record. In the text annotation pipeline, I applied a conservative probability threshold of 0.8 when converting model outputs into binary labels. To demonstrate that the results are not sensitive to this choice, I re-estimate the models using alternative thresholds of 0.7 and 0.9

(Models 1 and 2 in Table A.8). Similarly, I imposed a minimum threshold of 5 bills (one above the median) that a firm must lobby a legislator on when constructing lobbying ties to minimize inferring spurious relationships. I also re-estimate the models using alternative thresholds of 3 (one above the second quartile) and 10 (one above the third quartile) bills to show that the coefficients on discriminatory policies and import interests do not change substantively (Model 3 and 4 in Table A.8).

While setting a minimum threshold for the number of bills in constructing lobbying ties helps reduce the likelihood of inferring spurious connections, it does not capture the fact that firms are geographically embedded within specific constituencies and that legislators have stronger incentives to respond to firms located in their geographical constituencies (Yang 2024; Artés et al. 2024). To account for this, I measure firms' business presence across all 50 states using a text-as-data approach from their corporate filings.[20] The proxy measures of firms' business presence across 50 states allows me to separate firm-legislator-year observations into two groups: in-state groups if they share the same state, and out-of-state group if they do not. For in-state ties, the lobby-transmitted interests (based on Equation 1) receives an additional scaling factor equal to the square root of the number of times a firm mentions substantive business presence in a given state in a particular year, reflecting the intensity of firm-legislator co-location. Therefore, each type of firms' interests (export, import, FDI, and discriminatory policies) now consists of its in-state component and out-of-state counterpart. For out-of-state groups, the construction of lobby-transmitted interests remain unchanged.

Regressing legislators' anti-trade rhetoric on these two sets of firms' interests reveals that both in-state and out-of-state lobbying matter (Table A.9 in the Appendix). Similar to the main results, firms' concerns about discriminatory polices (import interests) transmitted through in-state ties predict more (less) anti-trade rhetoric. This is reassuring since this method of

20. I subset all sentences (N = 9,639,707) in firms' financial filings that mention any of the 50 U.S. states. However, mentioning a state by name does not necessarily mean a firm has substantive business presence, such as operating some facilities and employing local residents, in that state. To address this issue, I employ a similar text annotation pipeline described in the main empirical section. Depending on the specific states being mentioned, each sentence is annotated using the hypothesis, "The company has business operations (e.g., facilities, offices, or employees) in the state of {...}," separately for each state. In other words, each sentence is evaluated against potentially multiple hypotheses, producing binary labels that indicate whether the company discussed its business presence in each relevant state. Repeating this process for all sentences and aggregating the results allows me to construct firm–year–level proxies of business presence across all 50 states.

constructing lobbying ties reflects the theoretical importance of geographical co-location of firms and legislators without arbitrarily imposing any threshold of bills that connect them. What is interesting is that out-of-state ties produce coefficients twice as large as those from in-state ties for both discriminatory policy and import concerns. While this may suggest that out-of-state lobbying represents a distinct channel of influence not constrained by electoral geography (Fordham and McKeown 2003), future research could synthesize the rich geographical information from corporate filings and the network data from lobbying disclosures to develop more robust measures of lobbying ties.

7 Discussion and conclusion

This paper demonstrates that firm-level experiences of foreign discrimination play a critical and previously underappreciated role in shaping legislative rhetoric. By leveraging a novel text-as-data approach and constructing new measures of lobby-transmitted firm interests, I show that U.S. legislators become more critical of trade with China when lobbied by firms reporting policy discrimination. The substance of foreign discrimination and firms' concerns also changed over time: experiences of intellectual property disputes have become the new driver of trade criticism during the Trump and Biden administrations. In contrast, firms with global value chain (GVC) engagements, whether through export, import, or FDI, exert a weaker and asymmetric influence in countering anti-trade rhetoric.

This paper makes several theoretical interventions in the existing literature. First, I bring the host-government politics back to the international political economy (IPE) literature that places firms at the center of trade politics. By endogenizing firms interests as "the product of political strategies and manipulations" of the host governments(Lake 2009), I emphasize their "space and autonomy to act against foreign firms' interests" (Wellhausen 2015). I argue that the role of state power in IPE is likely to become even more prominent, as the state has dramatically expanded its role in recent years. Consider three developments in the global economy. First, the resurgence of geopolitical conflicts has compelled states to take supply chain resilience seriously and to recognize how trade and economic interdependence can be

weaponized by adversaries (Farrell and Newman 2019). Second, the race for technological leadership in emerging industries like artificial intelligence, characterized by extreme capital intensity and network effects, has prompted governments to devote substantial resources to industrial policy. Third, the rise of populism—partly fueled by dislocation from globalization—has created pressure for states to compensate "losers" and mitigate globalization's adverse effects, sometimes through protectionist policies. Taken together, these developments suggest a resurgence of state power in the global economy and underscore the need for a more interactive framework that incorporates how firms engage with both host and home governments.

Second, I identify conditions under which firms engaged in GVCs can fragment and mobilize anti-trade coalitions, contrary to the pro-trade expectations emphasized in the literature. I join Brutger (2024) in highlighting how firms, even domestic ones, can help monitor and enforce international trade law and advocate for the removal of market access barriers. My findings also echo the early work of Nollen and Quinn (1994), who noted that "fair trade" coalitions include liberal Members of Congress and those with strong financial backing from international businesses, and "are not simply protectionism in disguise."

Third, the finding that the China shock did not significantly promote anti-China trade rhetoric in Congress casts doubt on voter-driven models of the recent backlash against globalization. A common explanation of this backlash is that workers harmed by import competition reward or punish politicians at the ballot box, thereby pushing them toward protectionism. This assumption underlies much of the China shock-inspired studies. In contrast, this article highlights the role of lobbying by firms adversely impacted by international trade. It is consistent with the view that foreign policymakers are less responsive to public opinion or labor organizations and most influenced by business elites (Gilens and Page 2014). One implication is that one should be skeptical when politicians use the China shock and the induced job losses to justify the ongoing trade conflicts with China.

On the policy implications for U.S.-China trade relations, this study fills an important gap in understanding how business interests conditioned the unraveling of the bilateral engagement. It offers an alternative to two prevailing and overly simplified views. On the one hand, some suggest that "the business community has never wavered in its unvarnished support for

US engagement with China" (Shambaugh 2025). But this perspective overlooks the considerable variation in firms' preferences, both across and within sectoral lines. Furthermore, a static perspective of business power cannot explain the temporal dynamic of US-China trade relations. On the other hand, some attribute the recurrent trade conflicts to the idiosyncratic preferences of Trump. Yet this perspective overlooks that many of the business complaints and policymakers' doubts surrounding the utility of engagement surfaced much earlier, particularly during the Obama administration.

The implication of my study is that tensions in U.S.–China trade may be more structural and enduring unless the underlying grievances of firms are addressed. Yet given the current mistrust of liberal international institutions like the World Trade Organization (WTO), the prospects for resolving such grievances through rule-based mechanisms are likely to fall flat. As a result, bilateral trade conflicts may persist.

Methodologically, I propose a novel measurement strategy to capture granular firm preferences using large language models (LLMs) in a transparent, reproducible, and efficient manner (Spirling 2023). I further contribute an original empirical framework for linking firm lobbying to legislative rhetoric through lobbying network data (Kim and Kunisky 2020). These measurement and estimation strategies are flexible and can be readily applied to a wide range of comparative or international political economy questions concerning firm influence, lobbying, and elite behavior.

References

Artés, Joaquín, Aaron R. Kaufman, Brian K. Richter, and Jeffrey F. Timmons. 2024. "Are Firms Gerrymandered?" *American Political Science Review,* 1–21.

Autor, David H., David Dorn, Gordon H. Hanson, and Kaveh Majlesi. 2020. "Importing Political Polarization? The Electoral Consequences of Rising Trade Exposure" [in en]. *American Economic Review* 110, no. 10 (October): 3139–3183.

Baccini, Leonardo. 2019. "The Economics and Politics of Preferential Trade Agreements" [in eng]. Place: Palo Alto Publisher: Annual Reviews, *Annual review of political science* 22 (1): 75–92.

Baccini, Leonardo, Iain Osgood, and Stephen Weymouth. 2019. "The service economy: U.S. trade coalitions in an era of deindustrialization" [in en]. *The Review of International Organizations* 14, no. 2 (June): 261–296.

Baccini, Leonardo, Pablo M. Pinto, and Stephen Weymouth. 2017. "The Distributional Consequences of Preferential Trade Liberalization: Firm-Level Evidence" [in en]. *International Organization* 71 (2): 373–395.

Bai, Chong-en, Chang-Tai Hsieh, and Zheng Song. 2020. "Special Deals with Chinese Characteristics." _Eprint: https://doi.org/10.1086/707189, *NBER Macroeconomics Annual* 34:341–379.

Baldwin, Richard, Rebecca Freeman, and Angelos Theodorakopoulos. 2023. "Hidden Exposure: Measuing U.S. Supply Chain Reliance." In *Brookings Papers on Economic Activity.*

Baldwin, Robert, and Christopher Magee. 1998. *Is Trade Policy for Sale? Congressional Voting on Recent Trade Bills* [in en]. Technical report w6376. Cambridge, MA: National Bureau of Economic Research, January.

Ballard-Rosa, Cameron, Mashail A. Malik, Stephanie J. Rickard, and Kenneth Scheve. 2021. "The Economic Origins of Authoritarian Values: Evidence From Local Trade Shocks in the United Kingdom" [in en]. *Comparative Political Studies* 54, no. 13 (November): 2321–2353.

Beaulieu, Eugene. 2002. "Factor or Industry Cleavages in Trade Policy? An Empirical Analysis of the Stolper–Samuelson Theorem" [in en]. *Economics & Politics* 14, no. 2 (July): 99–131.

Brutger, Ryan. 2024. "Litigation for Sale: Private Firms and WTO Dispute Escalation" [in en]. *American Political Science Review* 118, no. 3 (August): 1204–1221.

Burnham, Michael. 2024. "Stance detection: a practical guide to classifying political beliefs in text." *Political Science Research and Methods* (September): 1–18.

Burns, William J. 2013. *The U.S.–China Closing Statements for U.S.–China Strategic and Economic Dialogue.* Publisher: U.S. Department of State, July.

Büthe, Tim, and Helen V. Milner. 2014. "Foreign Direct Investment and Institutional Diversity in Trade Agreements: Credibility, Commitment, and Economic Flows in the Developing World, 1971–2007" [in en]. *World Politics* 66, no. 1 (January): 88–122.

Campello, Daniela, and Francisco Urdinez. 2021. "Voter and Legislator Responses to Localized Trade Shocks from China in Brazil" [in en]. *Comparative Political Studies* 54, no. 7 (June): 1131–1162.

Cha, Victor D. 2023. "Collective Resilience: Deterring China's Weaponization of Economic Interdependence" [in en]. *International Security* 48, no. 1 (July): 91–124.

Chen, Frederick R., and Jian Xu. 2023. "Partners with Benefits: When Multinational Corporations Succeed in Authoritarian Courts" [in en]. *International Organization* 77 (1): 144–178.

Chilton, Adam S., Helen V. Milner, and Dustin Tingley. 2020. "Reciprocity and Public Opposition to Foreign Direct Investment" [in en]. *British Journal of Political Science* 50, no. 1 (January): 129–153.

Cinelli, Carlos, and Chad Hazlett. 2020. "Making Sense of Sensitivity: Extending Omitted Variable Bias" [in en]. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 82, no. 1 (February): 39–67.

Colantone, Italo, and Piero Stanig. 2018a. "Global Competition and Brexit" [in en]. *American Political Science Review* 112, no. 2 (May): 201–218.

———. 2018b. "The Trade Origins of Economic Nationalism: Import Competition and Voting Behavior in Western Europe" [in en]. *American Journal of Political Science* 62, no. 4 (October): 936–953.

Dagan, Ido, Oren Glickman, and Bernardo Magnini. 2006. "The PASCAL Recognising Textual Entailment Challenge." In *Machine Learning Challenges. Evaluating Predictive Uncertainty, Visual Object Classification, and Recognising Tectual Entailment,* edited by Joaquin Quiñonero-Candela, Ido Dagan, Bernardo Magnini, and Florence d'Alché-Buc, 3944:177–190. Berlin, Heidelberg: Springer Berlin Heidelberg.

Dür, Andreas. 2007. "Foreign Discrimination, Protection for Exporters, and U.S. Trade Liberalization" [in en]. *International Studies Quarterly* 51, no. 2 (June): 457–480.

Dür, Andreas, Leonardo Baccini, and Manfred Elsig. 2014. "The design of international trade agreements: Introducing a new dataset" [in en]. *The Review of International Organizations* 9, no. 3 (September): 353–375.

Farrell, Henry, and Abraham L Newman. 2019. "Weaponized Interdependence: How Global Economic Networks Shape State Coercion" [in en]. *International Security* 44 (1): 42–79.

Feigenbaum, James J., and Andrew B. Hall. 2015. "How Legislators Respond to Localized Economic Shocks: Evidence from Chinese Import Competition" [in en]. *The Journal of Politics* 77, no. 4 (October): 1012–1030.

Ferrara, Andreas, Patrick A. Testa, and Liyang Zhou. 2024. "New area- and population-based geographic crosswalks for U.S. counties and congressional districts, 1790–2020" [in en]. *Historical Methods: A Journal of Quantitative and Interdisciplinary History* 57, no. 2 (April): 67–79.

Fordham, Benjamin O., and Timothy J. McKeown. 2003. "Selection and Influence: Interest Groups and Congressional Voting on Trade Policy" [in en]. *International Organization* 57 (3): 519–549.

Garcia-Macia, Daniel, Siddharth Kothari, Yutong Li, and Yifan Tao. 2025. *Industrial Policy in China: Quantification and Impact on Misallocation.* IMF Working Paper WP/25/155. Washington, D.C.: International Monetary Fund, August.

Gelman, Andrew. 2008. "Scaling regression inputs by dividing by two standard deviations" [in en]. *Statistics in Medicine* 27, no. 15 (July): 2865–2873.

Gilens, Martin, and Benjamin I. Page. 2014. "Testing Theories of American Politics: Elites, Interest Groups, and Average Citizens" [in en]. *Perspectives on Politics* 12, no. 3 (September): 564–581.

Gourevitch, Peter Alexis. 1986. *Politics in hard times : comparative responses to international economic crises* [in eng]. ISBN: 0801419735 Place: Ithaca Publication Title: Politics in hard times : comparative responses to international economic crises Series: Cornell studies in political economy. Cornell University Press.

Guisinger, Alexandra. 2009. "Determining Trade Policy: Do Voters Hold Politicians Accountable?" [In en]. *International Organization* 63, no. 3 (July): 533–557.

Hall, Richard L., and Alan V. Deardorff. 2006. "Lobbying as Legislative Subsidy" [in en]. *American Political Science Review* 100, no. 1 (February): 69–84.

Hassan, Tarek A., Stephan Hollander, Aakash Kalyani, Laurence Van Lent, Markus Schwedeler, and Ahmed Tahoun. 2025. "Text as Data in Economic Analysis" [in en]. *Journal of Economic Perspectives* 39, no. 3 (August): 193–220.

Hiscox, Michael J. 2002. "Commerce, Coalitions, and Factor Mobility: Evidence from Congressional Votes on Trade Legislation" [in en]. *American Political Science Review* 96, no. 3 (September): 593–608.

Hoberg, Gerard, and S. Katie Moon. 2017. "Offshore activities and financial vs operational hedging" [in en]. *Journal of Financial Economics* 125, no. 2 (August): 217–244.

Hsueh, Roselyn. 2011. *China's Regulatory State: A New Strategy for Globalization* [in en]. 1st ed. Cornell Studies in Political Economy. Ithaca: Cornell University Press.

Judd, Nicholas, Dan Drinkard, Jeremy Carbaugh, and Lindsay Young. 2017. *congressional-record: A parser for the Congressional Record.*

Kim, In Song. 2017. "Political Cleavages within Industry: Firm-level Lobbying for Trade Liberalization" [in en]. *American Political Science Review* 111, no. 1 (February): 1–20.

Kim, In Song. 2018. "LobbyView: Firm-level Lobbying & Congressional Bills Database" [in en]. *Working paper available from https://web.mit.edu/insong/www/pdf/lobbyview.pdf.*

Kim, In Song, and Dmitriy Kunisky. 2020. "Mapping Political Communities: A Statistical Analysis of Lobbying Networks in Legislative Politics" [in en]. *Political Analysis* (November): 1–20.

Kim, In Song, Helen V Milner, Thomas Bernauer, Iain Osgood, Gabriele Spilker, and Dustin Tingley. 2019. "Firms and Global Value Chains: Identifying Firms' Multidimensional Trade Preferences" [in en]. *International Studies Quarterly* 63, no. 1 (March): 153–167.

Kim, In Song, and Iain Osgood. 2019. "Firms in Trade and Trade Politics" [in en]. *Annual Review of Political Science* 22, no. 1 (May): 399–417.

Kim, Sung Eun. 2018. "Media Bias against Foreign Firms as a Veiled Trade Barrier: Evidence from Chinese Newspapers" [in en]. *American Political Science Review* 112, no. 4 (November): 954–970.

Kim, Sung Eun, Rebecca L. Perlman, and Grace Zeng. 2025. "The politics of rejection: Explaining Chinese import refusals" [in en]. *American Journal of Political Science* 69, no. 2 (April): 438–454.

Kuk, John Seungmin, Deborah Seligsohn, and Jiakun Jack Zhang. 2018. "From Tiananmen to Outsourcing: the Effect of Rising Import Competition on Congressional Voting Towards China" [in en]. *Journal of Contemporary China* 27, no. 109 (January): 103–119.

Lake, David A. 2009. "Open economy politics: A critical review" [in en]. *The Review of International Organizations* 4, no. 3 (September): 219–244.

Lauderdale, Benjamin E., and Alexander Herzog. 2016. "Measuring Political Positions from Legislative Speech" [in en]. *Political Analysis* 24 (3): 374–394.

Laurer, Moritz, Wouter Van Atteveldt, Andreu Casas, and Kasper Welbers. 2024. "Less Annotating, More Classifying: Addressing the Data Scarcity Issue of Supervised Machine Learning with Deep Transfer Learning and BERT-NLI." *Political Analysis* 32, no. 1 (January): 84–100.

Lee, Jieun, and Iain Osgood. 2021. "Firms Fight Back: Production Networks and Corporate Opposition to the China Trade War." In *Geopolitics, Supply Chains, and International Relations in East Asia,* edited by Etel Solingen, pp. 153–172. Cambridge: Cambridge University Press.

Lim, Darren J., and Victor A. Ferguson. 2022. "Informal economic sanctions: the political economy of Chinese coercion during the THAAD dispute" [in en]. *Review of International Political Economy* 29, no. 5 (September): 1525–1548.

Loughran, Tim, and Bill Mcdonald. 2016. "Textual Analysis in Accounting and Finance: A Survey." *Journal of Accounting Research* 54, no. 4 (September): 1187–1230.

Mansfield, Edward D., and Diana C. Mutz. 2009. "Support for Free Trade: Self-Interest, Sociotropic Politics, and Out-Group Anxiety" [in en]. *International Organization* 63, no. 3 (July): 425–457.

McDonald, Bill. 2024. *The Notre Dame Software Repository for Accounting and Finance (SRAF).*

Menaldo, Victor, and Nicolas Wittstock. 2021. "Does technology transfer from the US to China harm American firms, workers, and consumers? A historical and analytic investigation" [in en]. *Economic and Political Studies* 9, no. 4 (October): 417–446.

Meyerrose, Anna M., and Sara Watson. 2024. "The Effects of Import Shocks, Electoral Institutions, and Radical Party Competition on Legislator Ideology: Evidence from France" [in en]. *British Journal of Political Science* (January): 1–26.

Milner, Helen V. 1988. *Resisting protectionism : global industries and the politics of international trade* [in eng]. Publication Title: Resisting protectionism : global industries and the politics of international trade. Princeton, N.J: Princeton University Press.

Milner, Helen V., and Dustin H. Tingley. 2011. "Who Supports Global Economic Engagement? The Sources of Preferences in American Foreign Economic Policy" [in en]. *International Organization* 65, no. 1 (January): 37–68.

Minnich, John. n.d. "Re-Innovation Nation: The Political Economy of Technology Transfer Policy in Post-WTO China." _Eprint: https://doi.org/10.1086/737174, *The Journal of Politics* 0 (ja): null.

Mullahy, John, and Edward C. Norton. 2024. "Why Transform Y ? The Pitfalls of Transformed Regressions with a Mass at Zero*" [in en]. *Oxford Bulletin of Economics and Statistics* 86, no. 2 (April): 417–447.

Mutz, Diana C., and Eunji Kim. 2017. "The Impact of In-group Favoritism on Trade Preferences" [in en]. *International Organization* 71 (4): 827–850.

Naughton, Barry. 2021. *The Rise of China's Industrial Policy: 1978 to 2020.* Technical report. Universidad Nacional Autónoma de México.

Nollen, Stanley D., and Dennis P. Quinn. 1994. "Free trade, fair trade, strategic trade, and protectionism in the U.S. Congress, 1987–88" [in en]. *International Organization* 48 (3): 491–525.

Osgood, Iain. 2016. "Differentiated Products, Divided Industries: Firm Preferences over Trade Liberalization" [in en]. *Economics & Politics* 28, no. 2 (July): 161–180.

———. 2018. "Globalizing the Supply Chain: Firm and Industrial Support for US Trade Agreements" [in en]. *International Organization* 72 (2): 455–484.

Pearson, Margaret M. 2005. "The Business of Governing Business in China: Institutions and Norms of the Emerging Regulatory State" [in en]. *World Politics* 57, no. 2 (January): 296–322.

PRC State Council. 2015. *Notice of the State Council on the Publication of \emphMade in China 2025.* Edited by Ben Murphy. Translated by Etcetera Language Group, Inc. Published: Website of the Central People's Government of the People's Republic of China, May.

Rodrik, Dani. 2010. "Making Room for China in the World Economy" [in en]. Publisher: American Economic Association, *American Economic Review* 100, no. 2 (May): 89–93.

Rogowski, Ronald. 1989. *Commerce and coalitions : how trade affects domestic political alignments* [in eng]. Princeton University Press.

Schnakenberg, Keith E. 2017. "Informational Lobbying and Legislative Voting" [in en]. *American Journal of Political Science* 61, no. 1 (January): 129–145.

Schwarz, Daniel, Denise Traber, and Kenneth Benoit. 2017. "Estimating Intra-Party Preferences: Comparing Speeches to Votes" [in en]. *Political Science Research and Methods* 5, no. 2 (April): 379–396.

Schweinberger, Tanja. 2022. "How promise breaking in trade rhetoric shapes attitudes toward bilateral US-China trade cooperation" [in en]. *Business and Politics* 24, no. 4 (December): 463–490.

Shambaugh, David L. 2025. *Breaking the engagement : how China won & lost America* [in eng]. ISBN: 9780197792421 Place: New York, NY Publication Title: Breaking the engagement : how China won & lost America. Oxford University Press.

Spirling, Arthur. 2023. "Why open-source generative AI models are an ethical way forward for science." *Nature (London)* 616 (7957): 413–413.

Stewart, Charles. 2021a. *House Standing Committee Assignments, 103rd - 117th Congress.* Version Number: V1 Published: Harvard Dataverse.

———. 2021b. *Senate Standing Committee Assignments, 103rd - 117th Congress.* Version Number: V1 Published: Harvard Dataverse.

Tan, Yeling. 2020. "Disaggregating "China, Inc.": The Hierarchical Politics of WTO Entry" [in en]. *Comparative Political Studies* 53, no. 13 (November): 2118–2152.

Tsai, Kellee S., and Barry Naughton. 2015. "Introduction: State Capitalism and the Chinese Economic Miracle" [in en]. In *State Capitalism, Institutional Adaptation, and the Chinese Miracle,* 1st ed., edited by Barry Naughton and Kellee S. Tsai, 1–24. Cambridge University Press, June.

USCBC. 2017. *Section 301 Investigation: China's Acts, Policies, and Practices Related to Technology Transfer, Intellectual Property and Innovation.* Published: Hearing submission to the Office of the United States Trade Representative, September.

USTR. 2018. *Findings of the Investigation into China's Acts, Policies, and Practices Related to Technology Transfer, Intellectual Property, and Innovation under Section 301 of the Trade Act of 1974.* Technical report. Washington, D.C.: Executive Office of the President, March.

Weiss, Jessica Chen, Panle Jia Barwick, Shanjun Li, and Jeremy L. Wallace. 2023. "Commercial Casualties: Political Boycotts and International Disputes" [in en]. *Journal of East Asian Studies* (October): 1–24.

Wellhausen, Rachel L. 2015. *The shield of nationality : when governments break contracts with foreign firms* [in eng]. Place: New York, NY Publication Title: The shield of nationality : when governments break contracts with foreign firms. Cambridge University Press.

Yang, Joonseok. 2024. "Why Compete for Firms? Electoral Effects of Corporate Headquarters Relocation" [in en]. *American Journal of Political Science* 68, no. 2 (April): 339–353.

Young, Alasdair R. 2016. "Not your parents' trade politics: the Transatlantic Trade and Investment Partnership negotiations" [in en]. *Review of International Political Economy* 23, no. 3 (May): 345–378.

Zhang, Hao. 2023. "Commerce, Coalitions, and Global Value Chains: Coordinated and Collective Lobbying on Trade" [in en]. *SSRN Electronic Journal.*

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Table A.1: Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Pctl. 25	Pctl. 75	Max
Rhetoric: oppose free trade	12040	0.57	3.7	0	0	0	117
Rhetoric: unfair trade practices	12040	0.94	6.3	0	0	0	283
Rhetoric: Technology theft	12040	0.32	2.8	0	0	0	113
Rhetoric: China as military threat	12040	0.091	1.2	0	0	0	53
Lobby interest: Export	12040	83	87	0	3.2	134	519
Lobby interest: Import	12040	24	34	0	0	32	266
Lobby interest: FDI	12040	139	143	0	5.2	222	885
Lobby interest: Any GVC interests	12040	246	255	0	8.2	389	1507
Lobby interest: Discriminatory polices	12040	6.8	10	0	0	9.5	77
Lobby interest: IP violation	12040	6.8	9.4	0	0	9.4	87
Party	12040						
... D	5888	49%					
... I	42	0%					
... R	6110	51%					
Chamber	12040						
... House	9778	81%					
... Senate	2262	19%					
DW-NOMINATE Dim 1	12040	0.052	0.43	-0.78	-0.37	0.45	0.94
Financed-related committees	12040	0.28	0.47	0	0	1	2
Foreign policy-related committees	12040	0.36	0.55	0	0	1	3
Tenure	12040	11	9.7	0	3	16	59
District competitiveness	9618	0.28	0.45	0	0	1	1
NAICS: Resources	9612	5.7	2	0	4.8	7	11
NAICS: Construction	9612	9.4	0.8	0	9.2	9.8	11
NAICS: Manufactur.	9612	10	0.96	0	9.7	10	12
NAICS: Trade	9612	11	0.84	0	11	11	12
NAICS: Information	9612	8.6	0.98	0	8.1	9.1	12
NAISC: Financial	9612	9.6	0.88	0	9.3	10	12
NAICS: Education/health	9612	11	0.84	0	11	11	12
NAICS: Professional	9612	10	0.97	0	10	11	12
NAICS: Leisure/hospitality	9612	10	0.83	0	10	11	12
% degree holders	9544	0.28	0.088	0.092	0.22	0.34	0.69
Unemployment rate	9557	0.06	0.023	0.02	0.044	0.073	0.18
Import exposure to China per US worker	9125	5.2	8	0.005	0.4	6.6	55

Table A.2: Alternative explanations for legislators' rhetoric on trade with China, 2001 to 2022.

	Oppose trade			
	(1)	(2)	(3)	(4)
Oppose trade (lagged)	0.256***	0.344***	0.154***	0.182***
	(0.040)	(0.051)	(0.023)	(0.026)
Export	0.001	0.001	0.002	
	(0.004)	(0.004)	(0.004)	
Import	-0.030***	-0.028***	-0.029***	
	(0.005)	(0.005)	(0.004)	
FDI	0.0002	-0.0001	0.0001	
	(0.002)	(0.003)	(0.003)	
Discriminatory policies	0.065***	0.063***	0.051***	
	(0.010)	(0.011)	(0.008)	
Emp. in natural resources (log)	-0.008			
	(0.047)			
Emp. in construction (log)	0.373			
	(0.283)			
Emp. in manufacturing (log)	-0.056			
	(0.091)			
Emp. in trade & transport (log)	-0.406			
	(0.592)			
Emp. in information (log)	0.002			
	(0.177)			
Emp. in finance & real estate (log)	-0.747*			
	(0.300)			
Emp. in professional services (log)	0.932**			
	(0.337)			
Emp. in education & health (log)	0.698+			
	(0.421)			
Emp. in leisure & hospitality (log)	-0.247			
	(0.403)			
College degree (%)	-3.04+			
	(1.78)			
Unemployment rate (%)	-3.20			
	(5.45)			
Exposure to China's import		-0.009		
		(0.011)		
China as military threat			0.473***	
			(0.114)	
Export (lagged)				0.010**
				(0.004)
Import (lagged)				-0.028***
				(0.004)
FDI (lagged)				-0.004
				(0.002)
Discriminatory policies (lagged)				0.030**
				(0.009)
# State	41	41	50	50
# Year	22	21	22	22
Observations	8,777	8,411	11,198	11,198
Chamber	House	House	Both	Both
Legislator-level controls	✓	✓	✓	✓
State fixed effects	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓

Notes. Coefficients from negative binomial models. Standard errors clustered by member ID are in parentheses. $^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table A.3: District electoral competitiveness, firms' lobby-transmitted interests and legislators' anti-trade rhetoric.

	Oppose trade	
	(1)	(2)
Oppose trade (lagged)	0.260***	0.265***
	(0.038)	(0.038)
Export	-0.002	-0.0006
	(0.004)	(0.005)
Import	-0.027***	-0.030***
	(0.005)	(0.005)
FDI	0.002	0.001
	(0.003)	(0.003)
Discriminatory policies	0.057***	0.063***
	(0.012)	(0.011)
Competitive district	0.021	
	(0.169)	
Export × Competitive district	0.010	
	(0.008)	
Import × Competitive district	-0.006	
	(0.006)	
FDI × Competitive district	-0.007	
	(0.005)	
Discriminatory policies × Competitive district	0.036+	
	(0.019)	
Competitive district (lagged)		0.186
		(0.177)
Export × Competitive district (lagged)		0.004
		(0.009)
Import × Competitive district (lagged)		0.001
		(0.006)
FDI × Competitive district (lagged)		-0.003
		(0.006)
Discriminatory policies × Competitive district (lagged)		0.015
		(0.020)
Observations	8,845	8,814
Legislator-level controls	✓	✓
State fixed effects (42)	✓	✓
Year fixed effects (22)	✓	✓

Notes. Coefficients from negative binomial models. Standard errors clustered by member ID are in parentheses. $^{+}p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A.4: Firm lobbying on policy discrimination and legislative rhetoric on unfair trade practices, by Presidency

	Unfair trade	
	(1)	(2)
Unfair trade (lagged)	0.102***	0.103***
	(0.014)	(0.015)
GVC interests × presidencyBush1	0.044	
	(0.036)	
GVC interests × presidencyBush2	-0.003***	
	(0.0005)	
GVC interests × presidencyObama1	-0.001*	
	(0.0005)	
GVC interests × presidencyObama2	0.001	
	(0.0008)	
GVC interests × presidencyTrump	-0.004***	
	(0.0009)	
GVC interests × presidencyBiden	-0.001	
	(0.002)	
Discriminatory policies × presidencyBush2	0.050+	0.045
	(0.028)	(0.029)
Discriminatory policies × presidencyObama1	0.089***	0.085***
	(0.012)	(0.012)
Discriminatory policies × presidencyObama2	0.067**	0.072**
	(0.022)	(0.023)
Discriminatory policies × presidencyTrump	0.057***	0.049***
	(0.011)	(0.012)
Discriminatory policies × presidencyBiden	-0.016	-0.039
	(0.042)	(0.051)
Export × presidencyBush1		0.010
		(0.109)
Export × presidencyBush2		-0.020
		(0.014)
Export × presidencyObama1		0.009
		(0.006)
Export × presidencyObama2		-0.008
		(0.006)
Export × presidencyTrump		-0.013*
		(0.006)
Export × presidencyBiden		-0.002
		(0.013)
Import × presidencyBush1		-0.056
		(0.301)
Import × presidencyBush2		0.027*
		(0.014)
Import × presidencyObama1		-0.036**
		(0.013)
Import × presidencyObama2		-0.005
		(0.012)
Import × presidencyTrump		-0.003
		(0.005)
Import × presidencyBiden		-0.030*
		(0.012)
FDI × presidencyBush1		0.060
		(0.057)
FDI × presidencyBush2		0.004
		(0.008)
FDI × presidencyObama1		-0.003
		(0.003)
FDI × presidencyObama2		0.007*
		(0.003)
FDI × presidencyTrump		0.001
		(0.005)
FDI × presidencyBiden		0.014*
		(0.006)
Observations	11,198	11,198
Legislator-level controls	✓	✓
State fixed effects (50)	✓	✓
presidency fixed effects (6)	✓	✓

Notes. Coefficients from negative binomial models. Standard errors clustered by member ID are in parentheses. $^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table A.5: Firm lobbying on IP violations and legislative rhetoric on IP theft, by Presidency

	Stealing US tech	
	(1)	(2)
Stealing US tech (lagged)	0.208***	0.212***
	(0.030)	(0.030)
GVC interests × presidencyBush1	-0.015	
	(0.066)	
GVC interests × presidencyBush2	-0.002+	
	(0.0010)	
GVC interests × presidencyObama1	0.001	
	(0.0008)	
GVC interests × presidencyObama2	0.001	
	(0.001)	
GVC interests × presidencyTrump	-0.004***	
	(0.0009)	
GVC interests × presidencyBiden	-0.012***	
	(0.004)	
IP violation × presidencyBush2	-0.009	-0.033
	(0.049)	(0.052)
IP violation × presidencyObama1	-0.078**	-0.074*
	(0.029)	(0.029)
IP violation × presidencyObama2	-0.043	-0.026
	(0.034)	(0.037)
IP violation × presidencyTrump	0.088***	0.092***
	(0.018)	(0.020)
IP violation × presidencyBiden	0.088*	0.113*
	(0.038)	(0.044)
Export × presidencyBush1		-0.206
		(0.170)
Export × presidencyBush2		-0.013
		(0.016)
Export × presidencyObama1		-0.0008
		(0.007)
Export × presidencyObama2		-0.027**
		(0.009)
Export × presidencyTrump		-0.026***
		(0.006)
Export × presidencyBiden		-0.030*
		(0.015)
Import × presidencyBush1		0.169
		(0.441)
Import × presidencyBush2		-0.024
		(0.017)
Import × presidencyObama1		-0.006
		(0.016)
Import × presidencyObama2		-0.0004
		(0.018)
Import × presidencyTrump		-0.005
		(0.006)
Import × presidencyBiden		-0.030+
		(0.018)
FDI × presidencyBush1		0.026
		(0.105)
FDI × presidencyBush2		0.008
		(0.009)
FDI × presidencyObama1		0.003
		(0.004)
FDI × presidencyObama2		0.019***
		(0.005)
FDI × presidencyTrump		0.008*
		(0.004)
FDI × presidencyBiden		0.005
		(0.010)
Observations	11,112	11,112
Legislator-level controls	✓	✓
State fixed effects (49)	✓	✓
presidency fixed effects (6)	✓	✓

Notes. Coefficients from negative binomial models. Standard errors clustered by member ID are in parentheses. $^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table A.6: OLS specification of firms' lobby-transmitted interests and legislators' rhetoric on trade with China, 2001 to 2022.

	Oppose trade (1)
Oppose trade (lagged)	0.349***
	(0.037)
Export	0.005^{+}
	(0.003)
Import	-0.012***
	(0.003)
FDI	-0.002
	(0.002)
Discriminatory policies	0.024^{+}
	(0.013)
Party: Independent	3.21
	(2.17)
Party: Republican	-0.504^{+}
	(0.270)
Chamber: Senate	0.646**
	(0.207)
Left-right ideology	0.563
	(0.365)
Finance committees	0.003
	(0.103)
Foreign affairs committees	0.013
	(0.084)
Tenure (years)	0.011*
	(0.005)
Observations	11,198
State fixed effects (50)	✓
Year fixed effects (22)	✓

Notes. Coefficients from OLS models. Standard errors clustered by member ID are in parentheses.
$^{+}p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A.7: Member- and year-fixed effects specification of firms' lobby-transmitted interests and legislators' rhetoric on trade with China, 2001 to 2022.

	Oppose trade (1)
Oppose trade (lagged)	0.020***
	(0.006)
Export	0.001
	(0.003)
Import	-0.011*
	(0.004)
FDI	-0.0009
	(0.002)
Discriminatory policies	0.027***
	(0.007)
Tenure (years)	-0.006
	(0.009)
Observations	5,426
Member ID fixed effects (473)	✓
Year fixed effects (22)	✓

Notes. Coefficients from negative binomial models. Standard errors clustered by member ID and state are in parentheses. $^{+}p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A.8: Alternative thresholds for constructing text-based measures and lobbying ties

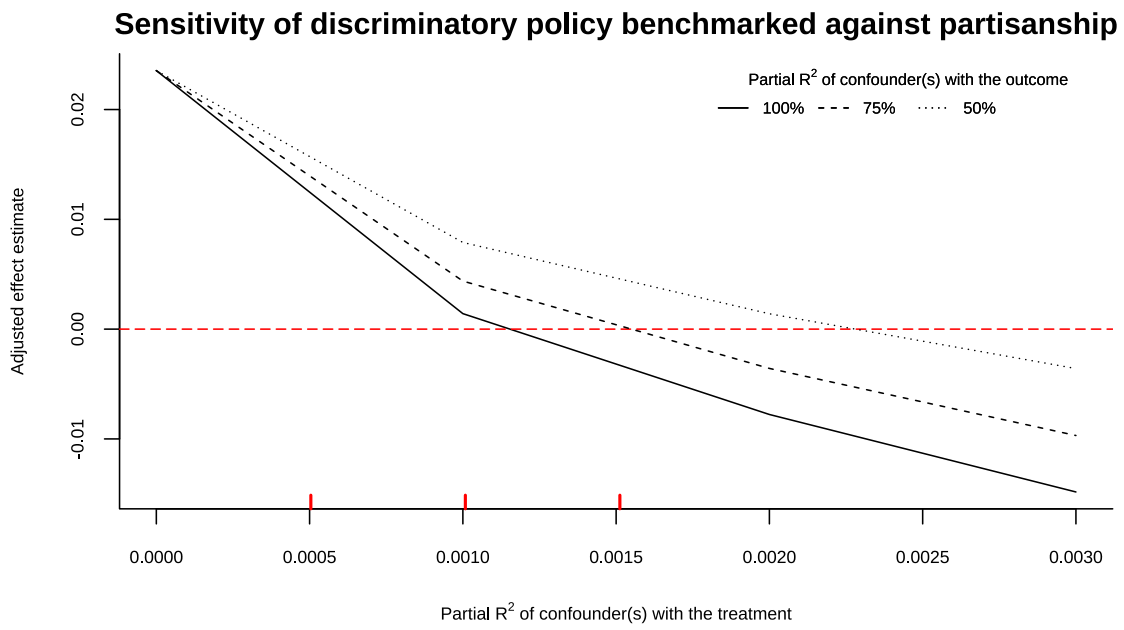
	Oppose trade			
	Label conversion prob = 0.7	Label conversion prob = 0.8	Min. bill threshold = 3	Min. bill threshold = 10
	(1)	(2)	(3)	(4)
Oppose trade (lagged)	0.164***	0.214***	0.184***	0.184***
	(0.022)	(0.030)	(0.025)	(0.025)
Export	0.002	0.008*	0.0006	0.010*
	(0.003)	(0.004)	(0.003)	(0.004)
Import	-0.028***	-0.034***	-0.026***	-0.027***
	(0.004)	(0.005)	(0.004)	(0.005)
FDI	0.0006	-0.003	0.0010	-0.005+
	(0.002)	(0.002)	(0.002)	(0.003)
Discriminatory policies	0.049***	0.062***	0.050***	0.043***
	(0.008)	(0.009)	(0.008)	(0.009)
Party: Independent	2.57*	2.78*	2.44*	2.52*
	(1.07)	(1.26)	(1.08)	(1.10)
Party: Republican	-0.842*	-1.07*	-0.929*	-0.943*
	(0.378)	(0.418)	(0.385)	(0.386)
Chamber: Senate	0.951***	0.905***	0.971***	0.943***
	(0.144)	(0.157)	(0.149)	(0.153)
Left-right ideology	1.22**	1.38**	1.27**	1.30**
	(0.472)	(0.523)	(0.485)	(0.478)
Finance committees	-0.104	-0.073	-0.116	-0.079
	(0.104)	(0.110)	(0.104)	(0.109)
Foreign affairs committees	0.310**	0.345**	0.282**	0.343**
	(0.105)	(0.112)	(0.107)	(0.110)
Tenure (years)	0.041***	0.044***	0.041***	0.041***
	(0.006)	(0.007)	(0.007)	(0.007)
# State	50	49	50	50
Observations	11,198	11,119	11,198	11,198
State fixed effects	✓	✓	✓	✓
Year fixed effects (22)	✓	✓	✓	✓

Notes. Coefficients from negative binomial models. Standard errors clustered by member ID are in parentheses. $^{+}p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A.9: In-state and out-of-state lobbying ties and legislators' rhetoric on trade with China

	Oppose trade		
	(1)	(2)	(3)
Oppose trade (lagged)	0.192***	0.185***	0.185***
	(0.026)	(0.026)	(0.026)
Export (In state)	0.003		0.003
	(0.002)		(0.002)
Import (In state)	-0.008**		-0.011***
	(0.003)		(0.003)
FDI (In state)	-0.0010		-0.0007
	(0.001)		(0.001)
Discriminatory policies (In state)	0.007		0.015***
	(0.004)		(0.004)
Export (Out of state)		0.001	0.002
		(0.003)	(0.003)
Import (Out of state)		-0.020***	-0.024***
		(0.004)	(0.004)
FDI (Out of state)		0.0005	0.0003
		(0.002)	(0.002)
Discriminatory policies (Out of state)		0.034***	0.043***
		(0.008)	(0.008)
Observations	11,198	11,198	11,198
Legislator-level controls	✓	✓	✓
State fixed effects (50)	✓	✓	✓
Year fixed effects (22)	✓	✓	✓

Notes. Coefficients from negative binomial models. Standard errors clustered by member ID are in parentheses. $^{+}p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.



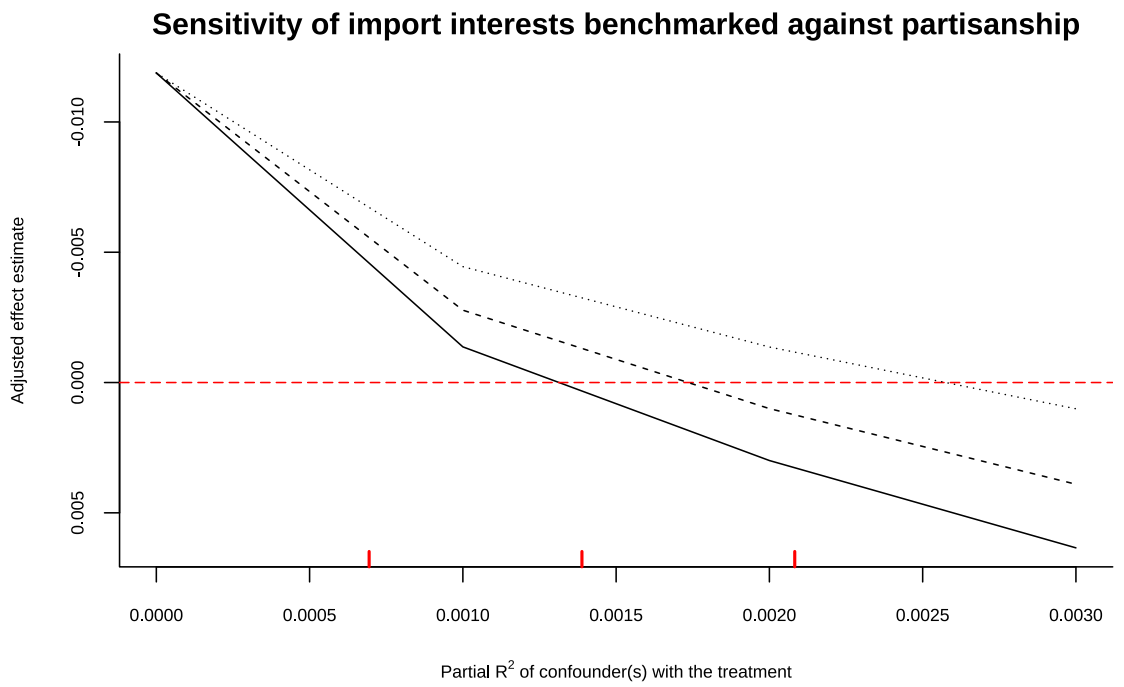


Figure A.1: Sensitivity analysis of policy discrimination and GVC interests with `sensemakr` in R (Cinelli and Hazlett 2020). The red ticks on the x-axis indicate bounds for confounders whose associations with the treatment is a multiple of that of partisanship (being a Republican).